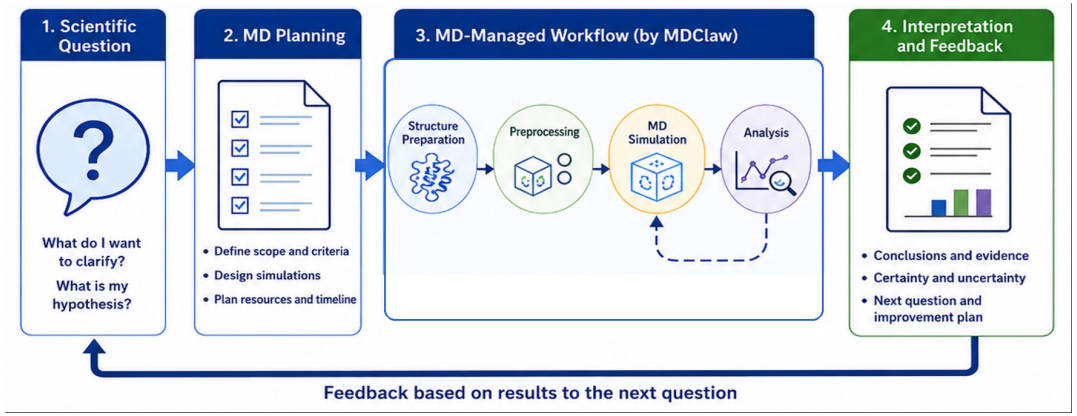
生命情報工学特論　第15回

AI Scientist研究とMDエージェント

研究の手順を記録し、
計算と評価を繰り返す



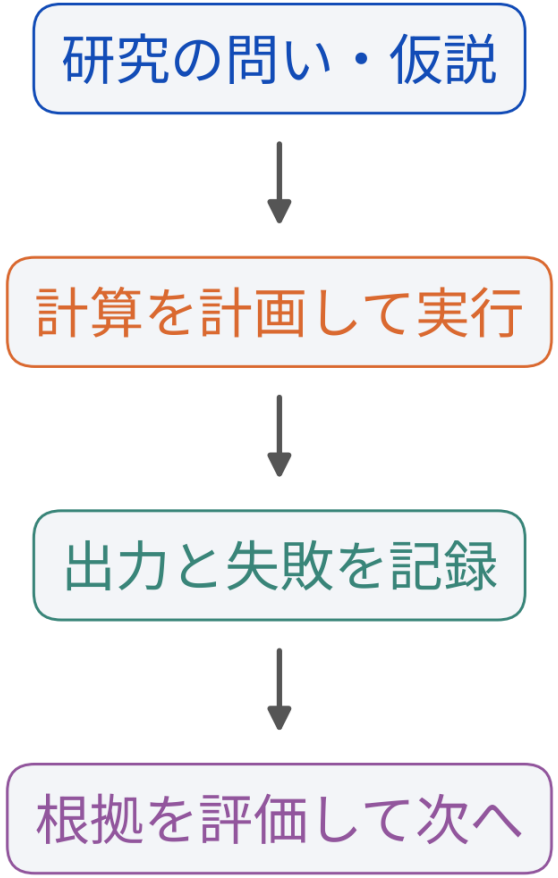
松永 康佑

埼玉大学 大学院理工学研究科
理化学研究所 計算科学研究センター

2027年1月18日

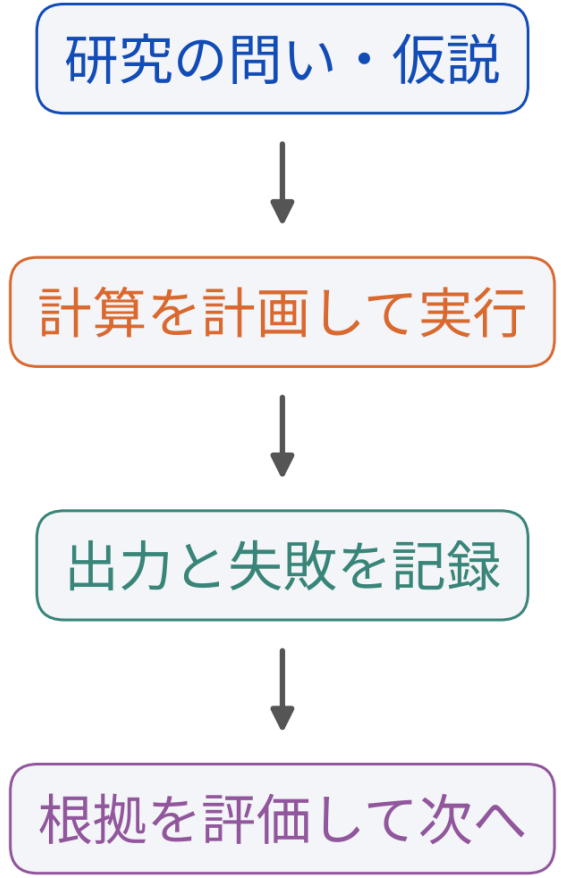
研究の手順を記録し、計算と評価を繰り返す

研究の手順を記録し、
計算と評価を繰り返す。
自然言語の仮説を計算手順へ落とし、出力を検証して次の計算へ進む。
データの量だけでなく、条件の統一・多様性・失敗の記録が研究の質を決める。



研究の自動化をデータと検証から考える

1　AI Scientistの提案と外部の検証
2　自動データ取得と計算条件の統一
3　転移学習とデータ量・多様性
4　MDClawのskills・CLI・計算履歴
5　教員デモ：入力から証拠までをたどる



Three AI Scientist papers published in May 2026:


Accelerated Article Preview

A multi-agent system for automating scientific discovery

nature

Accelerated Article Preview

Accelerating scientific discovery with Co-Scientist

nature

Accelerated Article Preview

An AI system to help scientists write expert-level empirical software

Paper	AI contribution	Validation loop	References
Robin **A multi-agent system for automating scientific discovery**	Literature-search and data-analysis agents propose hypotheses, experiments, and updated hypotheses.	**Lab-in-the-loop**: human scientists review candidates; assays and RNA-seq close the loop.	Ghareeb, A.E., Chang, B., Mitchener, L. et al. Nature (2026). Published online 19 May 2026. DOI: 10.1038/s41586-026-10652-y.
Co-Scientist **Accelerating scientific discovery with Co-Scientist**	A multi-agent system generates, critiques, ranks, and refines research hypotheses.	**Scientist-in-the-loop**: experts steer goals and select candidates for in vitro validation.	Gottweis, J., Weng, W.H., Daryin, A. et al. Nature (2026). Published online 19 May 2026. DOI: 10.1038/s41586-026-10644-y.
ERA **An AI system to help scientists write expert-level empirical software**	LLM plus tree search writes empirical scientific software against a quantitative score.	**Dataset-in-the-loop**: computational validations drive trial-and-error.	Aygun, E., Belyaeva, A., Comanici, G. et al. Nature (2026). Published online 19 May 2026. DOI: 10.1038/s41586-026-10658-6.

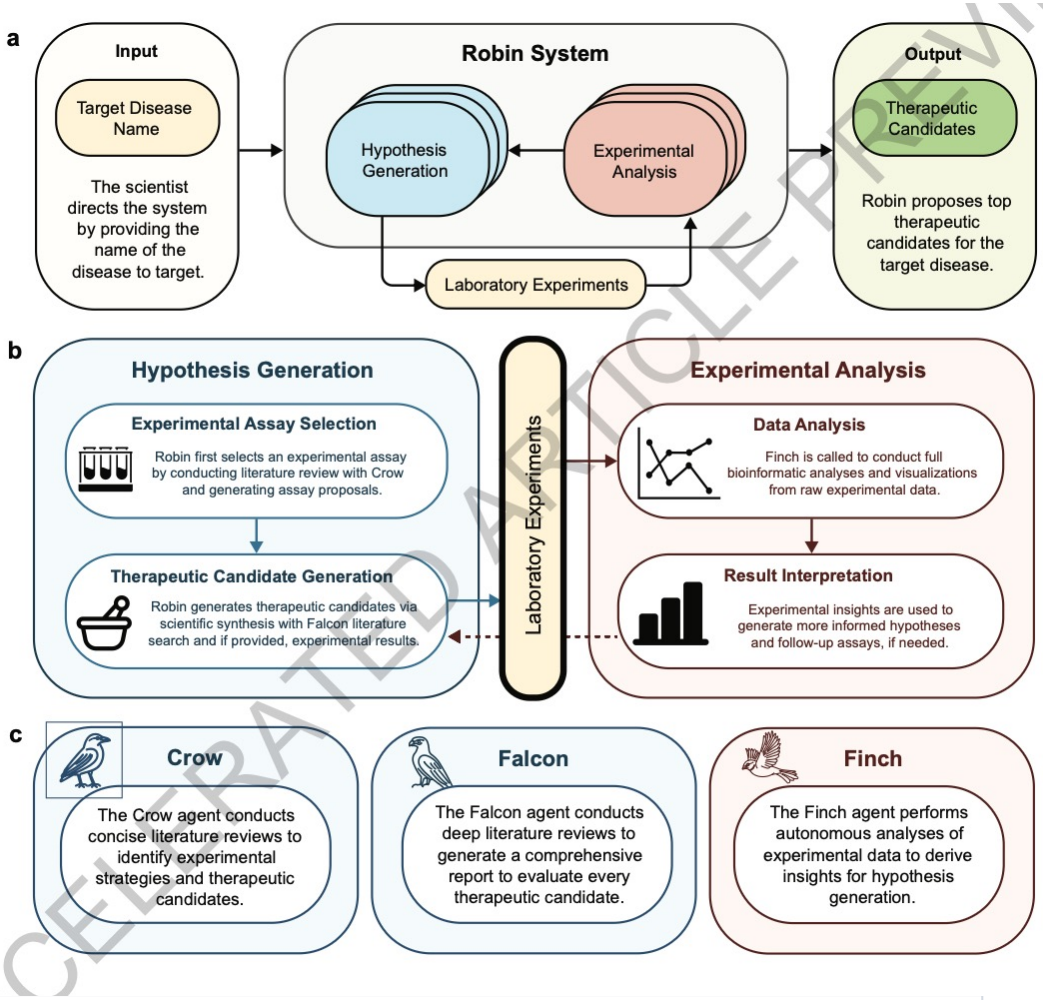
AI scales literature survey and scientific thinking; validation still the bottleneck

文献探索・仮説生成・コード作成を支援する系でも、実験・計算・専門家による評価が必要。2026年5月の原著3報。

Robin

Ghareeb, A.E., Chang, B., Mitchener, L. et al. Nature (2026). Published online 19 May 2026. DOI: 10.1038/s41586-026-10652-y.

- Consists of two literature search agents, *Crow* and *Falcon*, and *Finch*, which analyzes experimental data and related information.
- For the target disease, it generates hypotheses, selects top therapeutic candidates, analyzes the resulting experimental data, and updates the hypotheses.



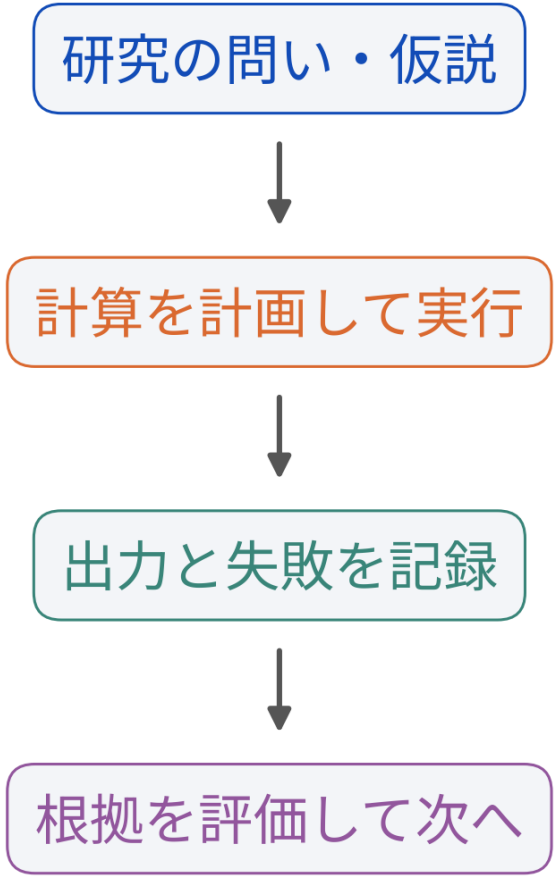
ACCELERATED ARTICLE PREVIEW

AI can propose hypotheses;
the scientific loop still needs an outside check: expert, lab, or physics(simulation).

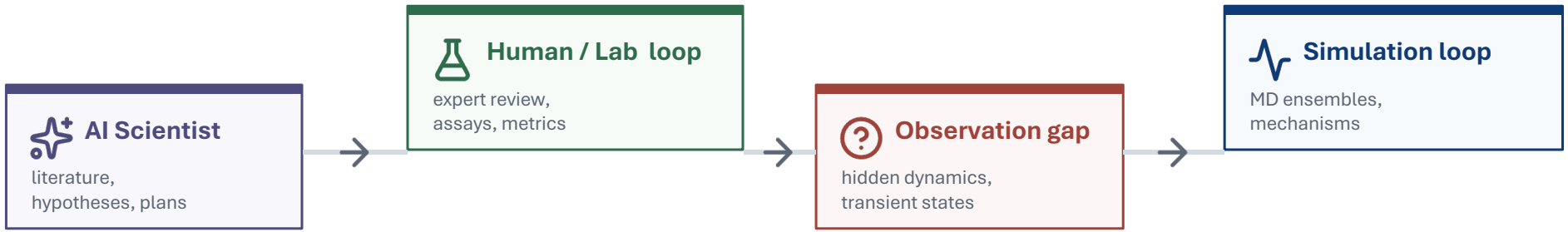
Robinは文献探索と実験データ解析を組み合わせる。実験系での検証を経て仮説を更新する。Ghareeb et al., Nature (2026)。

もっともらしい説明を、検証できる仮説へ変える

例：「変異で安定になる」を、そのまま計算の入力にはできない。
何を安定性と呼ぶか、何と比べるか、温度や溶媒は何かを決める。
仮説から観測量・比較条件・判定方法へ落とす段階で、研究者の判断が必要になる。
計算が成功したことと、仮説が支持されたことを分ける。

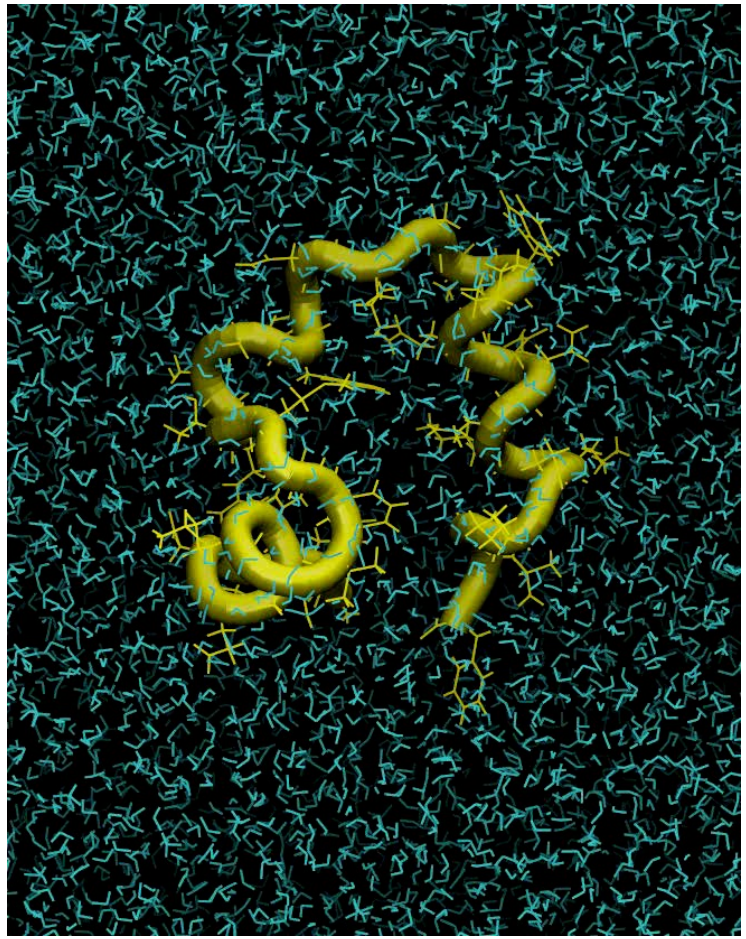


AI Scientists need simulation-in-the-loop when experiments cannot observe enough.



実験で試しにくい条件を計算で調べ、仮説と観測可能な量を結ぶ。計算モデル自体の検証も必要。

Molecular Dynamics (MD) simulation

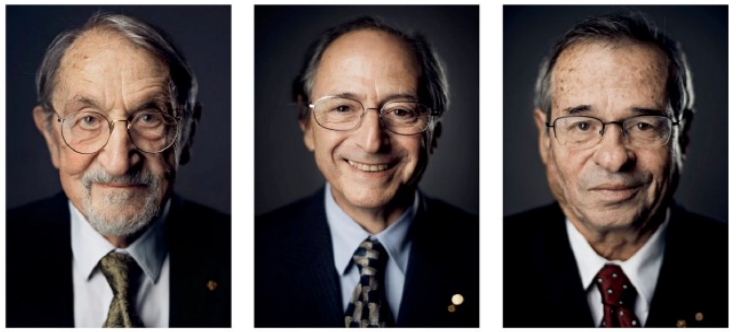


MD simulation solves Newtonian equation to sample molecular structures and dynamical properties of target molecules

Newtons' 2nd law

$$F = ma$$

The Nobel Prize in Chemistry 2013



© Nobel Media AB. Photo: A. Mahmoud
Martin Karplus
Prize share: 1/3

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Michael Levitt
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Arieh Warshel
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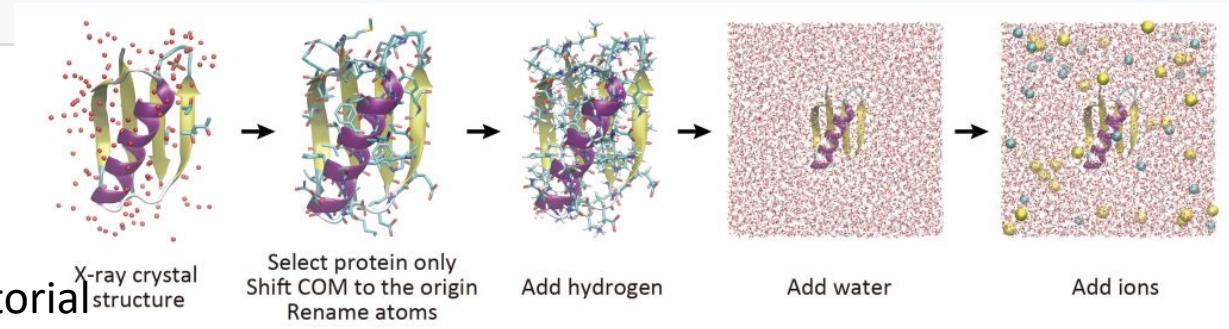
nobelprize.org

MD quantifies molecular mechanisms that cannot be directly observed in experiments by simulating atomic-scale motions and interactions.

MD pipelines are hard to automate with if–then rules.

MD automation is difficult because the protocol depends on molecular context, not a fully enumerable rule tree.

- System setup is not one deterministic command; each molecule brings different preparation choices.
- Mutations, PTMs, cofactors, ligands, membranes, and protonation states create many case-specific branches.
- Force-field selection, missing atoms/residues, charge assignment, and solvation choices often require expert judgment.
- If–then rules cannot reliably cover all exceptions without becoming brittle or incomplete.
- Simulation-in-the-loop MD needs an agent that can inspect inputs, validate results, and revise the protocol.

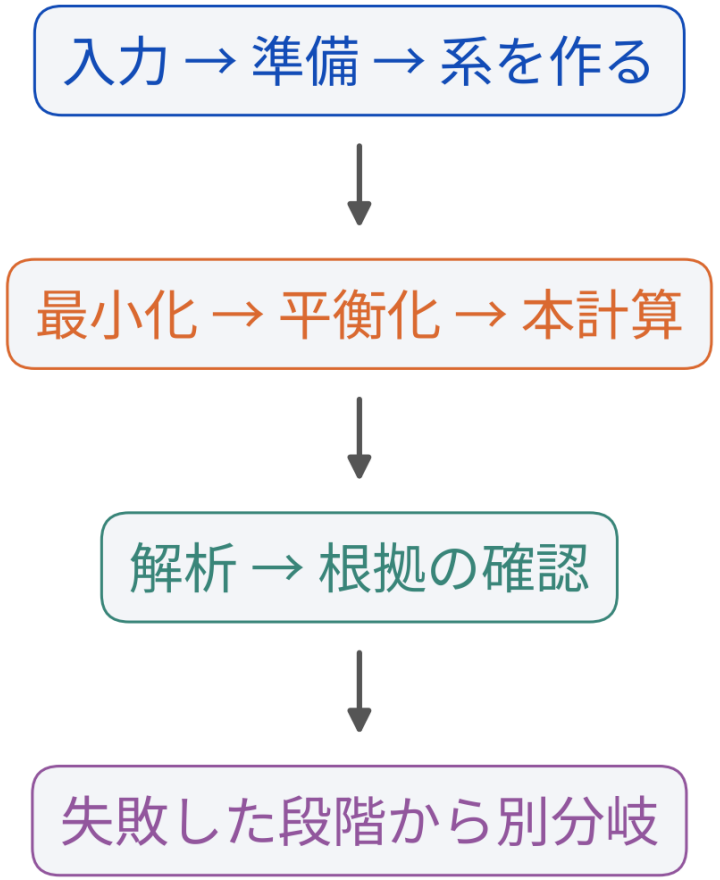


From GENESIS tutorial

MDの自動化では、化学状態・モデル・入力ファイルの選択を一貫して記録することが重要。

計算データを増やすには、条件も同時に揃える

異なる構造を同じ規則で準備し、同じ観測量を取り出す。
構造ID、配列、プロトン化状態、力場、水モデル、温度、乱数、計算長を保存する。
成功したものだけを残すと、難しい系が消えた偏ったデータ集合になる。失敗理由も残す。



Use case 1

抗体(Nanobody)の網羅的なシミュレーション (SAbDab dataset)



Training data generation

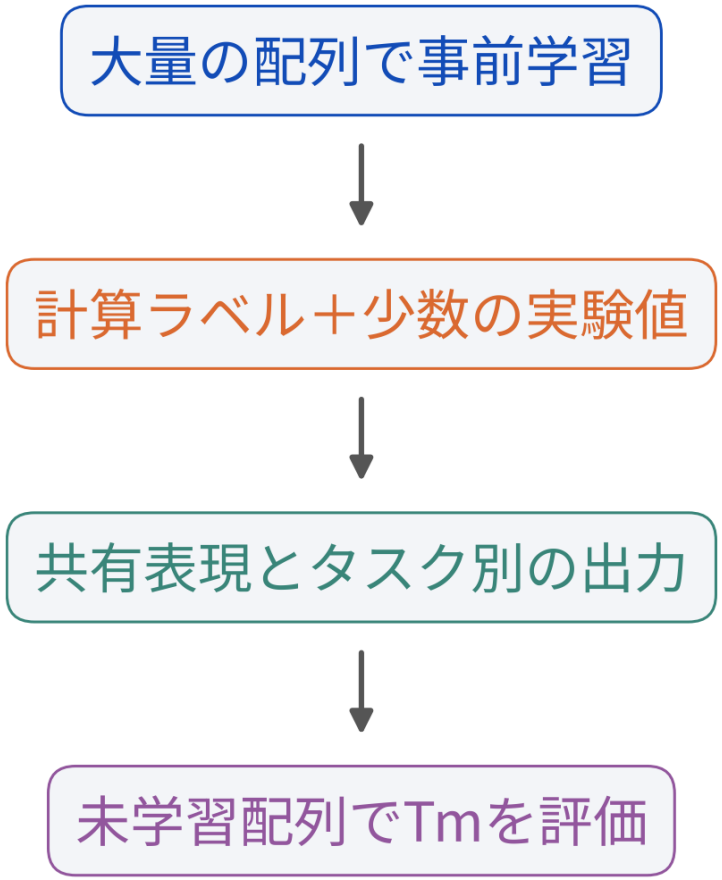
Prompt: Please perform 100 ns MD simulations for all the nanobodies in the table.

pdb	Hchain	Lchain	model	antigen_chain	antigen_type	antigen_het_name	antigen_name
8g2y	N	NA	0	A | B	protein | protein	NA | NA	minig alpha s/q chimera | guanine nucleotide-binding protein g(i)/g(s)/g(t) subunitbeta-1
8bb7	D	NA	0	B	protein	NA	plexin-b1
8bb7	C	NA	0	A	protein	NA	plexin-b1
8bf4	D	NA	0	F	protein	NA	plexin-b1
8bf4	E	NA	0	F	protein	NA	plexin-b1
8bf4	A	NA	0	C	protein	NA	plexin-b1
8bf4	B	NA	0	C	protein	NA	plexin-b1
7xk9	B	NA	0	A	protein	NA	endolysin,beta-2 adrenergic receptor
7xka	B	NA	0	A	protein	NA	endolysin,beta-2 adrenergic receptor
8flq	N	NA	0	A | G | B	protein | protein | protein	NA | NA | NA	guanine nucleotide-binding protein g(s) subunit alphaisoforms short | guanine nucleotide-binding prote
8flr	N	NA	0	A | G | B	protein | protein | protein	NA | NA | NA	guanine nucleotide-binding protein g(s) subunit alphaisoforms short | guanine nucleotide-binding prote
8fls	N	NA	0	A | B	protein | protein	NA | NA	guanine nucleotide-binding protein g(s) subunit alphaisoforms short | guanine nucleotide-binding prote
8flt	N	NA	0	A | B	protein | protein	NA | NA	guanine nucleotide-binding protein g(s) subunit alphaisoforms short | guanine nucleotide-binding prote
8flu	N	NA	0	A | B	protein | protein	NA | NA	guanine nucleotide-binding protein g(s) subunit alphaisoforms short | guanine nucleotide-binding prote
8sbb	B	NA	0	A	protein	NA	alkane 1-monooxygenase
7yit	D	NA	0	A	protein	NA	kappa-type opioid receptor
8fu6	N	NA	0	A | B	protein | protein	NA | NA	guanine nucleotide-binding protein g(s) subunit alphaisoforms short | guanine nucleotide-binding prote
8ont	B	NA	0	A	protein	NA	nramp related aluminium transporter
7uia	B	NA	0	A	protein	NA	neurotoxin type e
7uia	E	NA	0	D	protein	NA	neurotoxin type e
8cxr	E	NA	0	A	protein	NA	phospho-n-acetylmuramoyl-pentapeptide-transferase
8cxr	G	NA	0	C	protein	NA	phospho-n-acetylmuramoyl-pentapeptide-transferase
8cxr	F	NA	0	B	protein	NA	phospho-n-acetylmuramoyl-pentapeptide-transferase
8cxr	H	NA	0	D	protein	NA	phospho-n-acetylmuramoyl-pentapeptide-transferase
8f8v	A	NA	0	NA	NA	NA	NA
8f8v	B	NA	0	NA	NA	NA	NA
8f8w	C	NA	0	B	protein	NA	afucosylated igg1 fragment
8f8w	D	NA	0	A	protein	NA	afucosylated igg1 fragment
8f8w	G	NA	0	F	protein	NA	afucosylated igg1 fragment
8f8w	H	NA	0	E	protein	NA	afucosylated igg1 fragment
8f8x	C	NA	0	B	protein	NA	uncharacterized protein dkfzp686c11235
8f8x	D	NA	0	A	protein	NA	uncharacterized protein dkfzp686c11235
8i2g	N	NA	0	A | G | B	protein | protein | protein	NA | NA | NA	chimera of guanine nucleotide-binding protein g(s) subunitalpha and guanine nucleotide-binding protein
8en1	C	NA	0	A	protein	NA	gii.4 p domain

網羅的なMDで計算ラベルを作るという利用例。依頼文だけでなく対象一覧と条件・完了基準を定める。

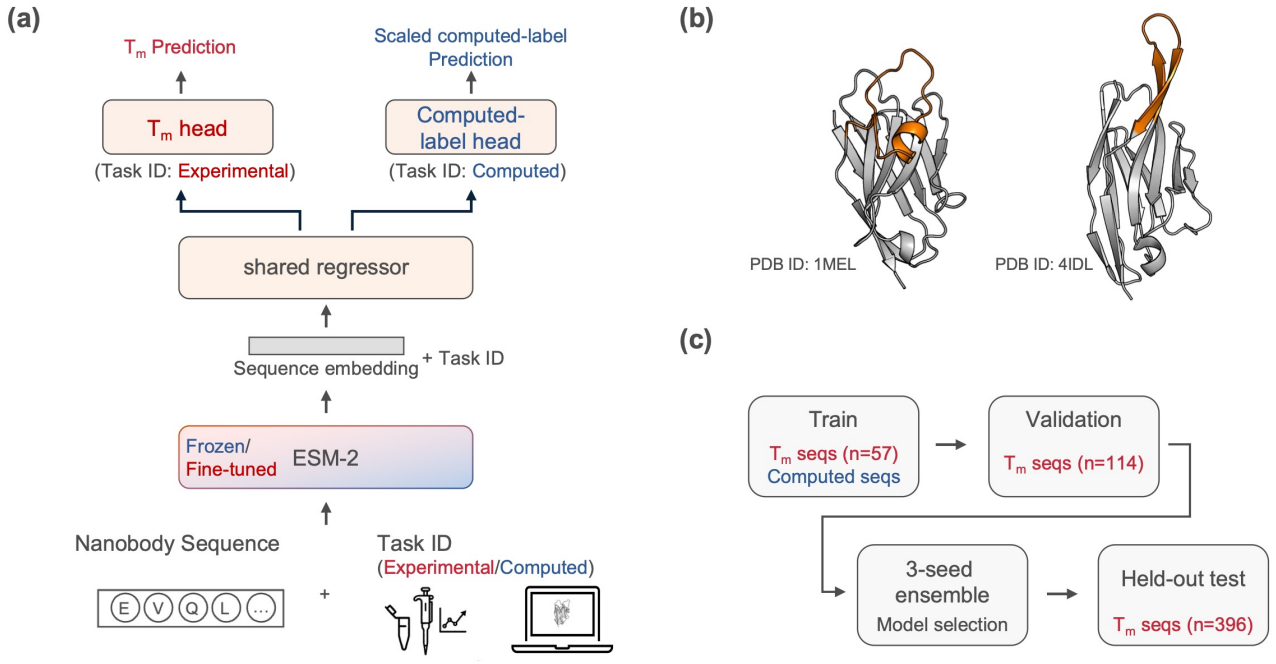
転移学習では、共有できる表現を計算データで育てる

タンパク質言語モデルが配列の表現を作る。
少数の実験Tmと、計算による安定性の補助ラベルを別の出力で学習する。
計算値を実験値と同一視するのではなく、共通の表現を介して実験予測を助ける。
最終評価には学習に使っていない実験配列を使う。



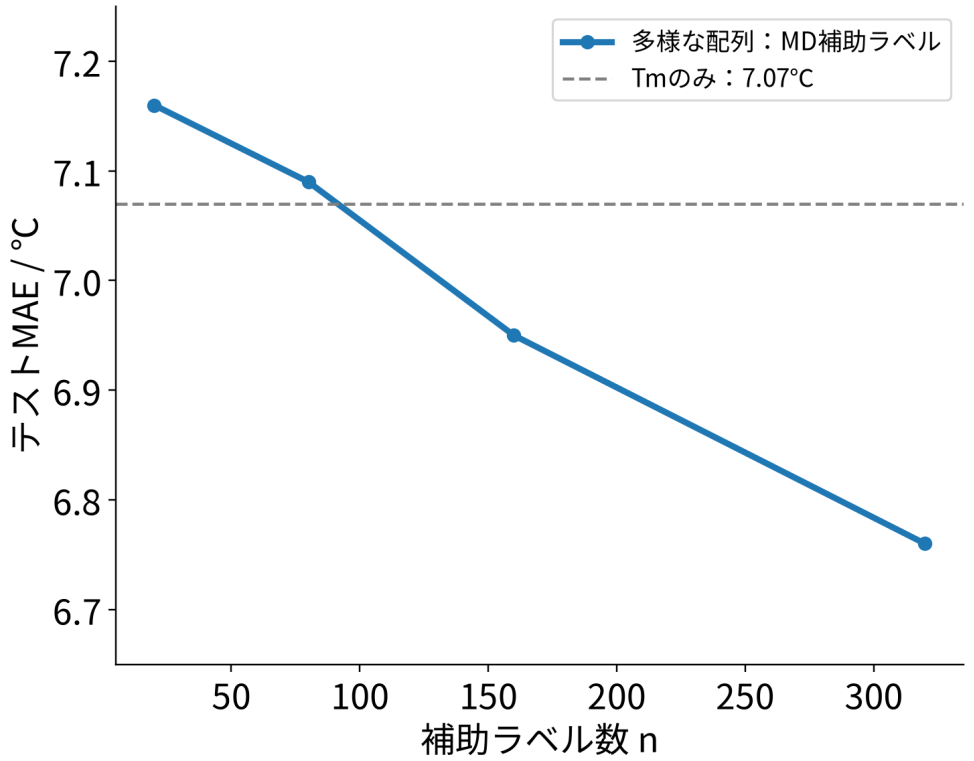
村上さんの研究：ナノボディの融解温度を予測する

Tmは熱変性の中点に対応する温度。FEPのΔΔGやMDの天然接触の保持率とは別の量。
実験Tm：学習57配列、モデル選択114配列、最終評価396配列。
計算ラベルを補助タスクとして、共有したESM2表現を学習する。
Murakami et al., bioRxiv (2026), DOI:10.64898/2026.07.30.741744。



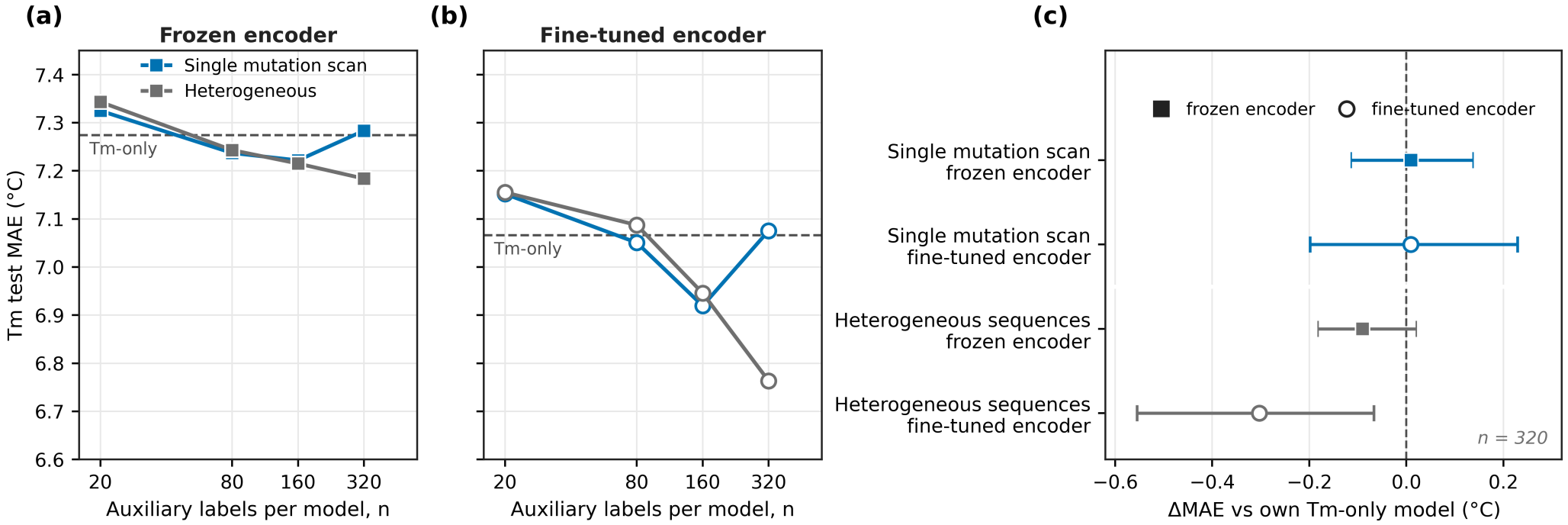
同じ計算量でも、どの配列を計算するかが効く

同じMD由来のラベルでも、少数構造の変異走査と、多様なナノボディ集合を比較する。
多様な集合では、ラベル数20→320で微調整モデルのMAEが7.16→6.76°Cへ下がった。
変異走査では単調な改善にならなかった。量だけでなく、覆う配列の広さが重要。
原稿Fig. 2の条件。右の再描画は本文の数値に基づく。



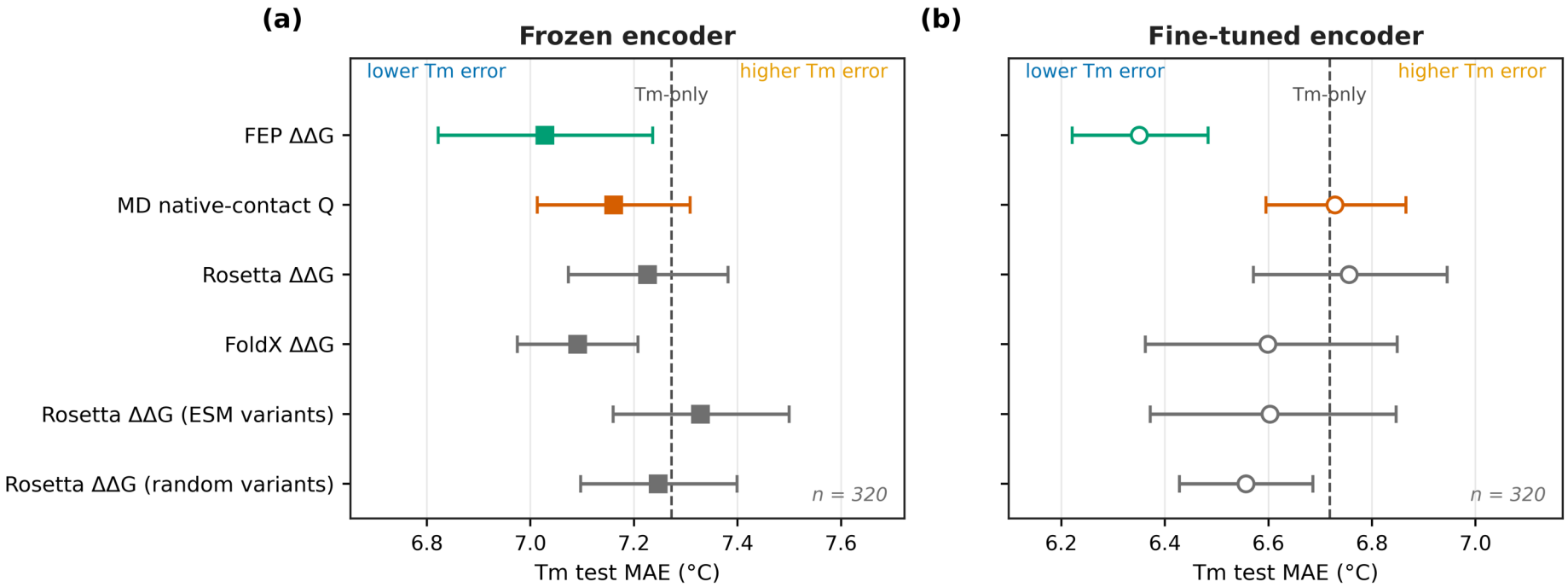
原著Fig. 2：データ設計によって増量の効果が違う

左・中央：ラベル数と、凍結／微調整したモデルの誤差。
右：同じラベル数320で、それぞれのTmのみの基準からの変化。
誤差が低いほどよい。異なる学習条件の基準値を混ぜない。
この範囲の結果から、無限に増やせば改善すると外挿しない。



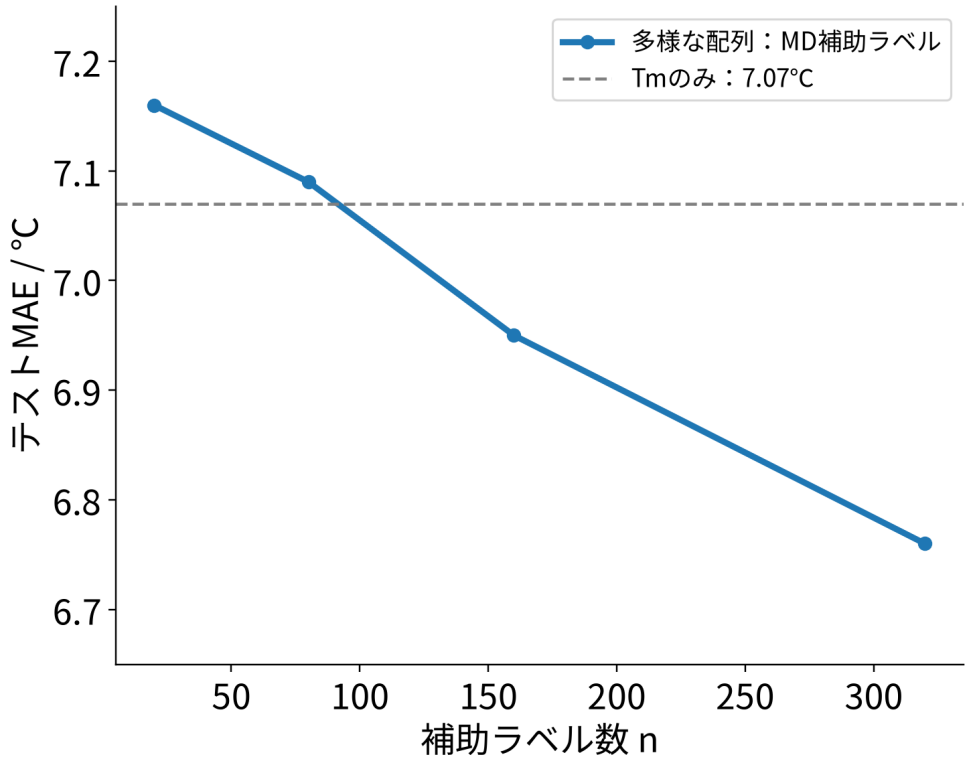
原著Fig. 3：何を計算するかも予測に影響する

同じ変異配列に対し、FEP・MD接触率・Rosetta・FoldXの補助ラベルを比較。
この比較ではFEP補助の微調整モデルのMAEが6.35°C。Tmのみは6.72°C。
「この予測に有用だった」ことと、「実験ΔΔGを最も正確に再現した」ことは別。



スケーリングの議論では、量・質・評価を分ける

データ数Nを増やすと誤差が減る経験的な関係は、対象・範囲・学習条件に依存する。
重複や偏りの強いデータを増やしても、未知の配列への誤差は減らないことがある。
類縁配列や条件が訓練とテストをまたぐ影響も点検する。
自動化は、比較可能なデータを狙って増やすために使う。



Our Goal

Our MD agent under development: MDClaw

Upstream questions

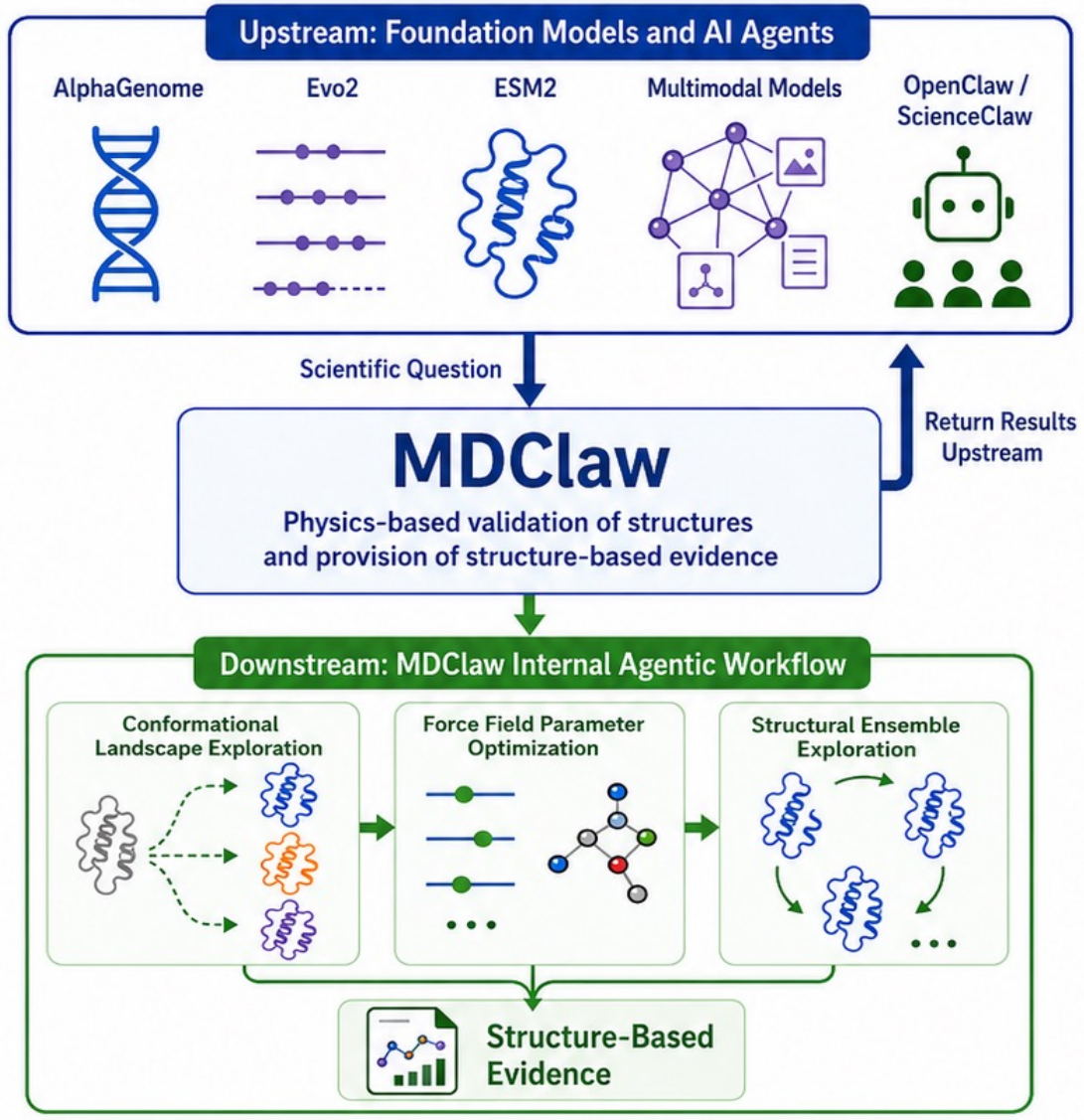
AlphaGenome, Evo2, gene language models, protein language models, and RIKEN AGIS foundation models generate scientific hypotheses.

MDClaw's role

Turn those questions into recorded MD plans, simulations, analyses, and evidence.

Return upstream

Send back structure-based evidence, uncertainty, and the next question for the AI system or scientist.



Overview of MDClaw

- MD agent for structure- and interaction-level validation that is difficult to verify experimentally.
- In biomolecular MD, it **autonomously carries out the chain from scientific question to validation-oriented MD planning**, system preparation, MD execution, analysis, and feedback.
- Implemented as a **Skills + Command Lines (CLI)-based system**, allowing it to run through different harnesses such as Pi, OpenCode, Claude Code, Codex, and OpenClaw.
- Supports validation across diverse system-generation scenarios, including mutations, modifications, protonation states, force-field file selection/customization, and initial-structure generation with Boltz-2 or BioEmu.
- Github repo https://github.com/matsunagalab/mdclaw

MDClawはskills・CLI・DAGを組み合わせる研究基盤。紹介図は2026年6月発表、役割分担は10月11日のREADMEでも確認。

MDClaw is created by agent skills and CLIs, and the skills tell the agent what to do; CLIs do the work and record the results.

Functions covered by agent skills and CLIs

md-study
question -> plan

md-prepare
source/prep/solv/topo

md-equilibration
staged relaxation

md-production
segments + restart

md-analyze
trajectory evidence

hpc-run
SLURM execution

skillが手順と判断基準を与え、CLIが操作を実行し、記録を残す。自然言語の説明と機械が読む出力を対応させる。

HOW AN AGENT CALLS MD TOOLS

Example of agent skills (markdown files)

md-prepare

source/prep/solv/topo

https://github.com/matsunagalab/mdclaw/blob/main/skills/md-prepare/SKILL.md

name	md-prepare
description	Molecular dynamics simulation preparation using MDClaw CLI tools. Covers structure acquisition, protein/nucleic/ligand selection, structure cleaning, solvation, and topology generation.

MD Prepare

You are a computational biophysics expert helping users set up MD simulations using the MDClaw CLI tools.

Read `skills/common/preamble.md`, `skills/common/tool-output.md`, `skills/common/defaults.md`, `skills/common/node-cli-patterns.md`, `skills/common/run-loop.md`, `skills/common/autonomous-checklist.md`, and `skills/common/guardrail-codes.md` before acting.

`skills/common/run-loop.md` is the per-step loop: call `mdclaw plan_next --job-dir <job_dir>` to learn the next node type, tool, and concrete parent IDs; let `create_node` auto-resolve the parent; then run the tool with node context. Use the IDs it returns, never literal example IDs.

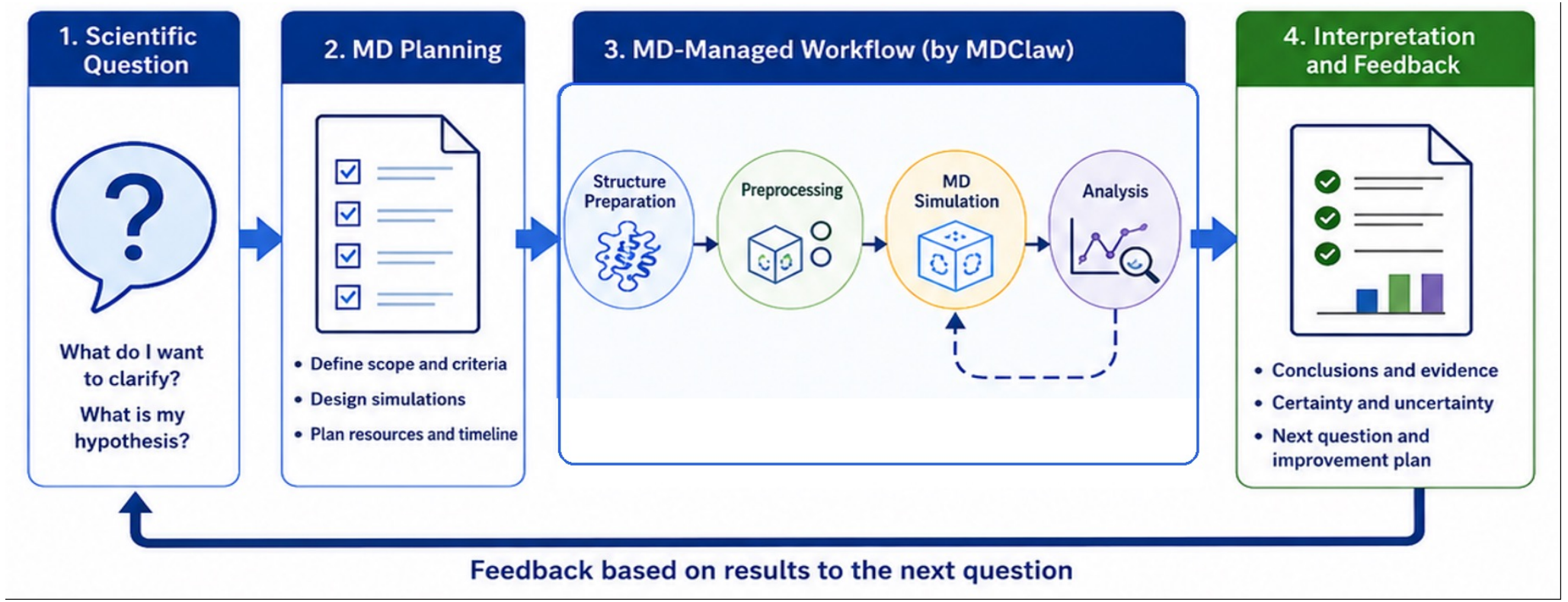
Defaults — Source of Truth

This project uses **ff19SB + OPC** as the modern explicit-water default (Amber Manual 2024 recommendation), NOT the legacy `ff14SB + tip3p` combination commonly seen in AMBER tutorials and training data. The pairing is enforced by guardrails — `ff19SB + tip3p` is rejected as a structured error (code `forcefield_water_blocked`).

Default solvation mode is **explicit solvent**. Unless the user explicitly requests implicit solvent, no-solvent/vacuum topology, or a membrane workflow, run `prepare_complex` → `solvate_structure` → `build_amber_system`. Topology tools must consume the completed DAG parent artifact (`solvated_pdb` for explicit/membrane, `merged_pdb` for implicit/vacuum); never pass a raw/manual PDB file directly into topology generation.

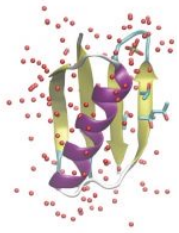
エージェントは手順書を読み、引数を指定してツールを呼ぶ。戻り値を読んで、次の操作や失敗への対応を判断する。

Pipeline of MDClaw

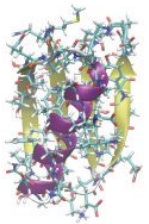


入力構造から準備・水和・トポロジー・最小化・平衡化・本計算・解析へ、成果物を受け渡す。

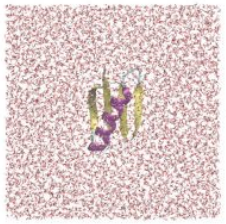
Directed Acyclic Graph (DAG)-based MD workflow is durable, auditable, and flexible (1/2)



X-ray crystal structure

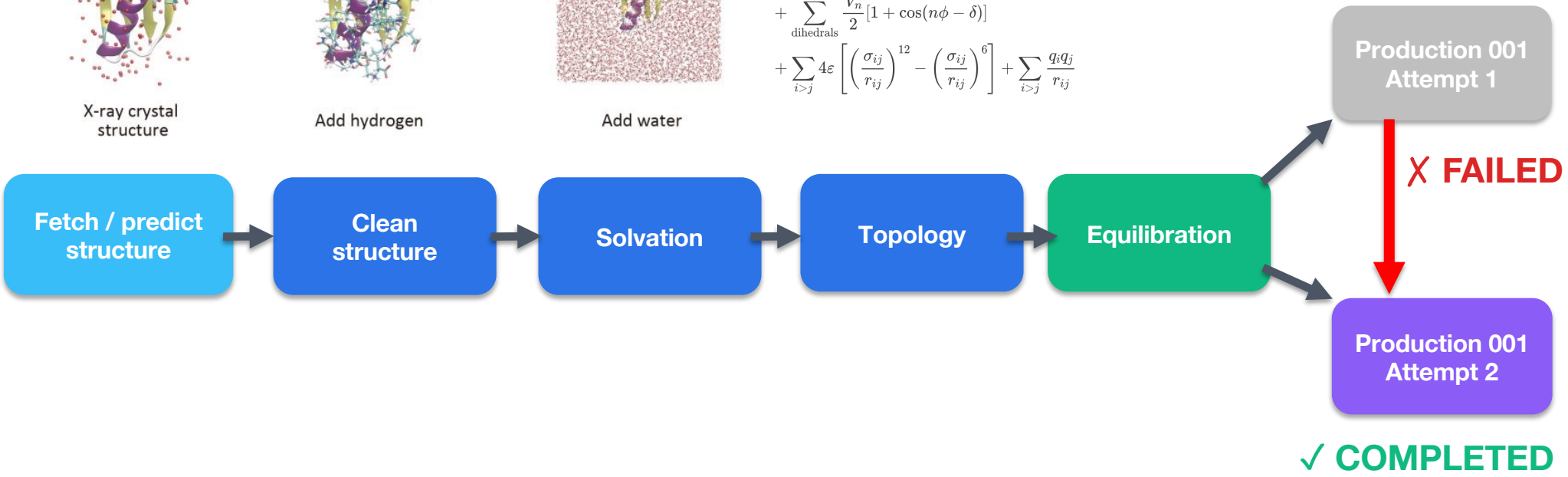


Add hydrogen



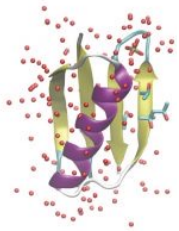
Add water

$$U = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_a(\theta - \theta_0)^2 + \sum_{\text{dihedrals}} \frac{V_n}{2}[1 + \cos(n\phi - \delta)] + \sum_{i>j} 4\varepsilon \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right] + \sum_{i>j} \frac{q_i q_j}{r_{ij}}$$

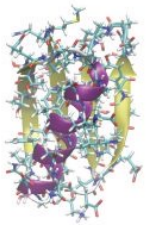


DAGは循環のない有向グラフ。失敗した実行と新しい試行を別ノードに残し、経緯を追えるようにする。

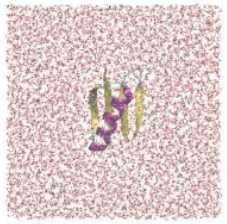
DAG-based MD workflow is durable, auditable, and flexible (2/2)



X-ray crystal structure

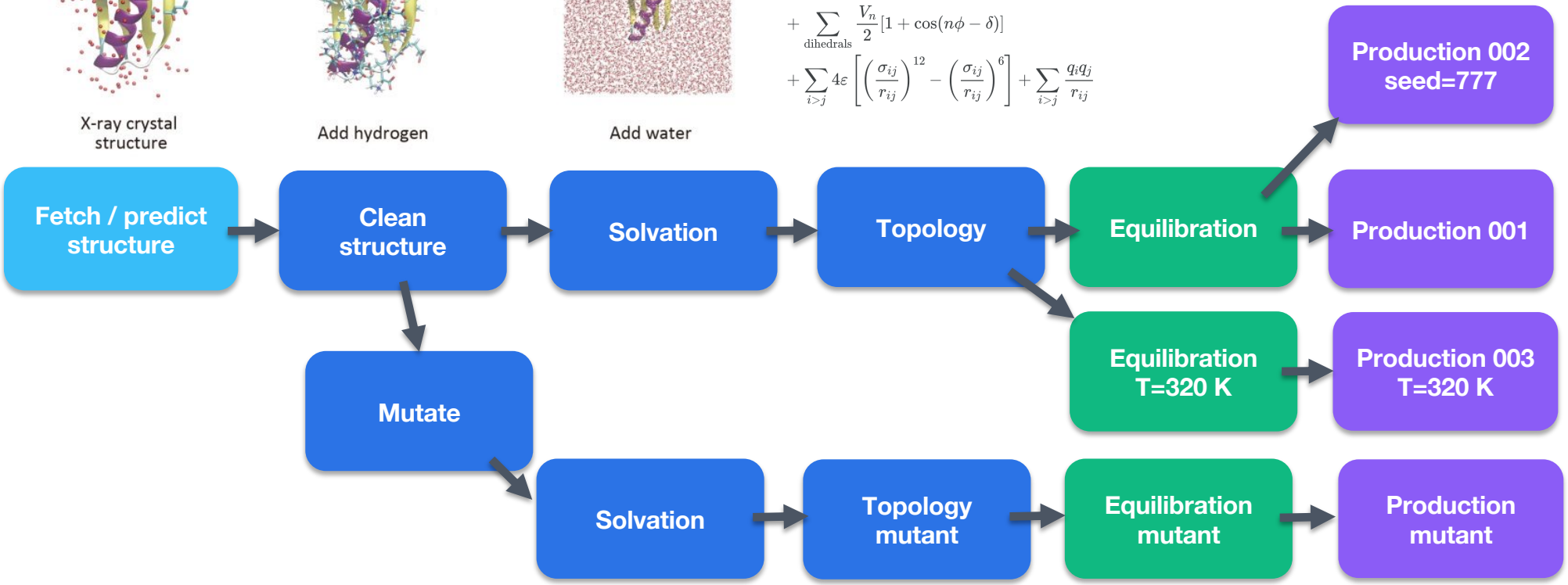


Add hydrogen



Add water

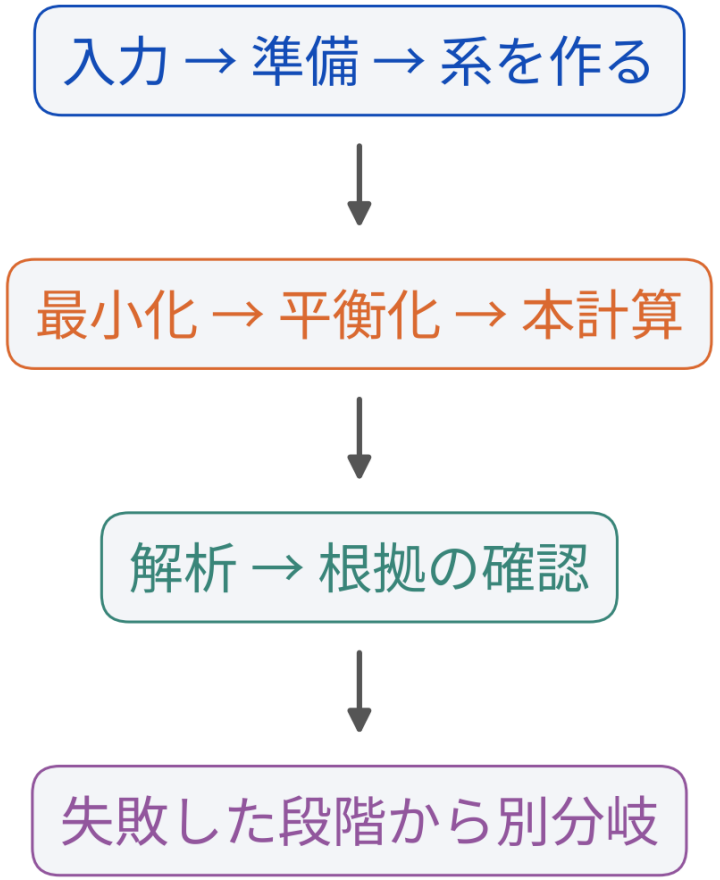
$$U = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_a(\theta - \theta_0)^2 + \sum_{\text{dihedrals}} \frac{V_n}{2}[1 + \cos(n\phi - \delta)] + \sum_{i>j} 4\varepsilon \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right] + \sum_{i>j} \frac{q_i q_j}{r_{ij}}$$



乱数、温度、変異などを変えた比較は別の分岐として保存する。共通の入力と変更した条件を区別する。

再実行では、完了済みの根拠を使って続きへ進む

inspect_jobで状態と成果物を読む。explain_nodeで次の段階の入力条件を確認する。
trace_failureで失敗した段階と理由を追う。
ファイルが存在するだけで完了とせず、正しい入力に対応した出力かを検査する。
2026年10月11日のローカルREADMEにある操作の役割。



CLI stops risky chemistry choices instead of quietly continuing.

Structured guardrail result

- field
- message
- severity
- actual / expected
- suggested_fix
- stable code

```
code: forcefield_water_blocked
actual: ff19SB + tip3p
expected: ff19SB + OPC
```

Topology contract

- prep supplies chemistry and records
- topo resolves force fields and parameters
- topo emits an OpenMM XML triple
- eq/prod/analyze consume DAG output files

```
system.system.xml
system.topology.pdb
system.state.xml
```

Result: agents can react to clear error codes, humans can check decisions, and doubtful systems do not quietly enter production MD.

電荷・結合・残基・水モデルなど、化学的な前提を点検する。入力の不整合を放置したまま本計算へ進めない。

MDPrepBench: Evaluating MD agent's abilities for preparing MD systems

What is measured

Can the agent turn messy public structural inputs into minimizable, reloadable MD systems?

Does it preserve requested chemistry: ligands, ions, PTMs, protonation, membranes, solvent model?

Does it submit clear records and evidence, not just a plausible narrative?

Submission contract

- manifest.json
- metrics.json
- provenance.json
- evidence_report.json
- prepared_structure.pdb
- minimized_structure.pdb
- minimization_report.json
- OpenMM XML triple

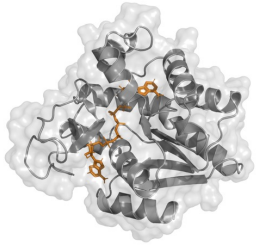
Scoring checks

- required and forbidden files
- JSON schema and file integrity
- component and residue-state rescans
- RMSD / candidate selection checks
- OpenMM system load and energy rescan
- minimization report and structure rescan

準備の評価では、再読込・エネルギー再計算・要求された化学状態の保持を点検する。最小化成功だけでは不十分。

Prepared structures are inspectable, not only scorable.

P02: chain A + AP5



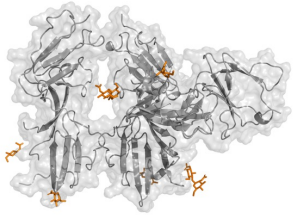
AP5 ligand retained; solvent hidden; protein shown with translucent surface.

P06: calmodulin + Ca2+



Calcium ions near the protein are visible as yellow spheres.

P14: glycoprotein



Glycan-like non-standard residues are emphasized in orange.

P17: DNA duplex



Prepared nucleic-acid system shown without solvent or periodic-box clutter.

生成された構造は数値評価だけでなく、成分・配置・欠損・意図しない変化を目で点検する。

Results of MDPrepBench

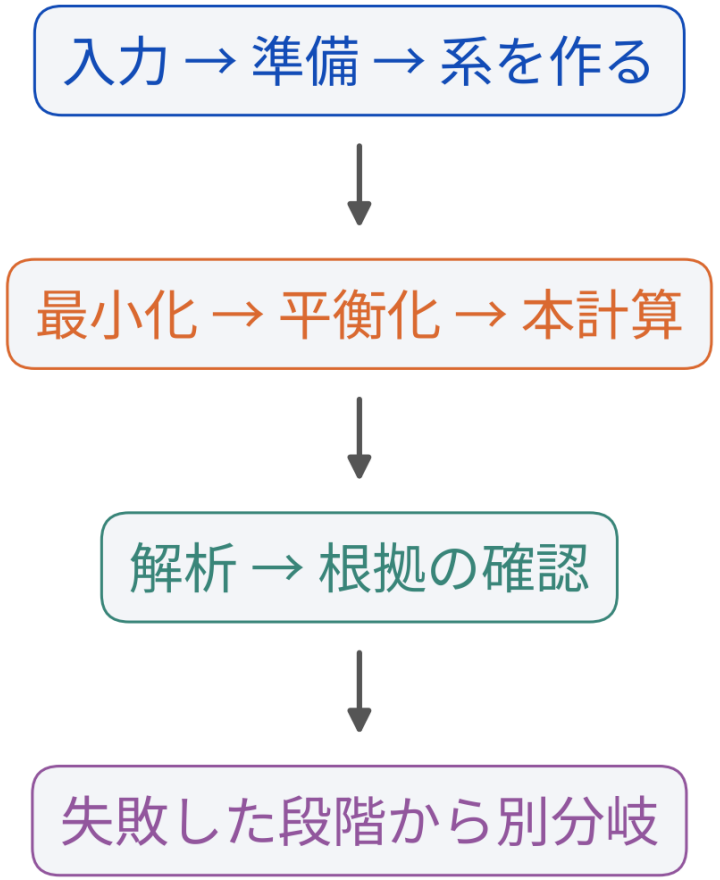
model / harness	skills+CLIs	CLIs-only	free
Haiku 4.5 / claude-code	22/25	7/25	5/25
DeepSeek-v4-flash / pi	20/25	7/25	13/25

- **In this 25-task comparison: haiku falls from skills+CLI to CLI-only to free (passes 22→7→5).**
- CLI alone adds little: CLI-only (7/25) and free (5/25) are close; the value is in the skill layer.

2026年6月発表時の25課題による比較。現在の版・別の課題集合の性能とは区別する。

ベンチマーク結果は、条件と失敗例まで読む

課題集合、入力、モデル版、制限時間、計算環境、採点基準を揃える。
途中で停止した実行や、提出物がない実行を完了した評価と混ぜない。
成功率に加え、どの化学状態や段階で失敗するかを調べる。
過去の25課題の点数から、現在の任意の系の成功率は予測できない。

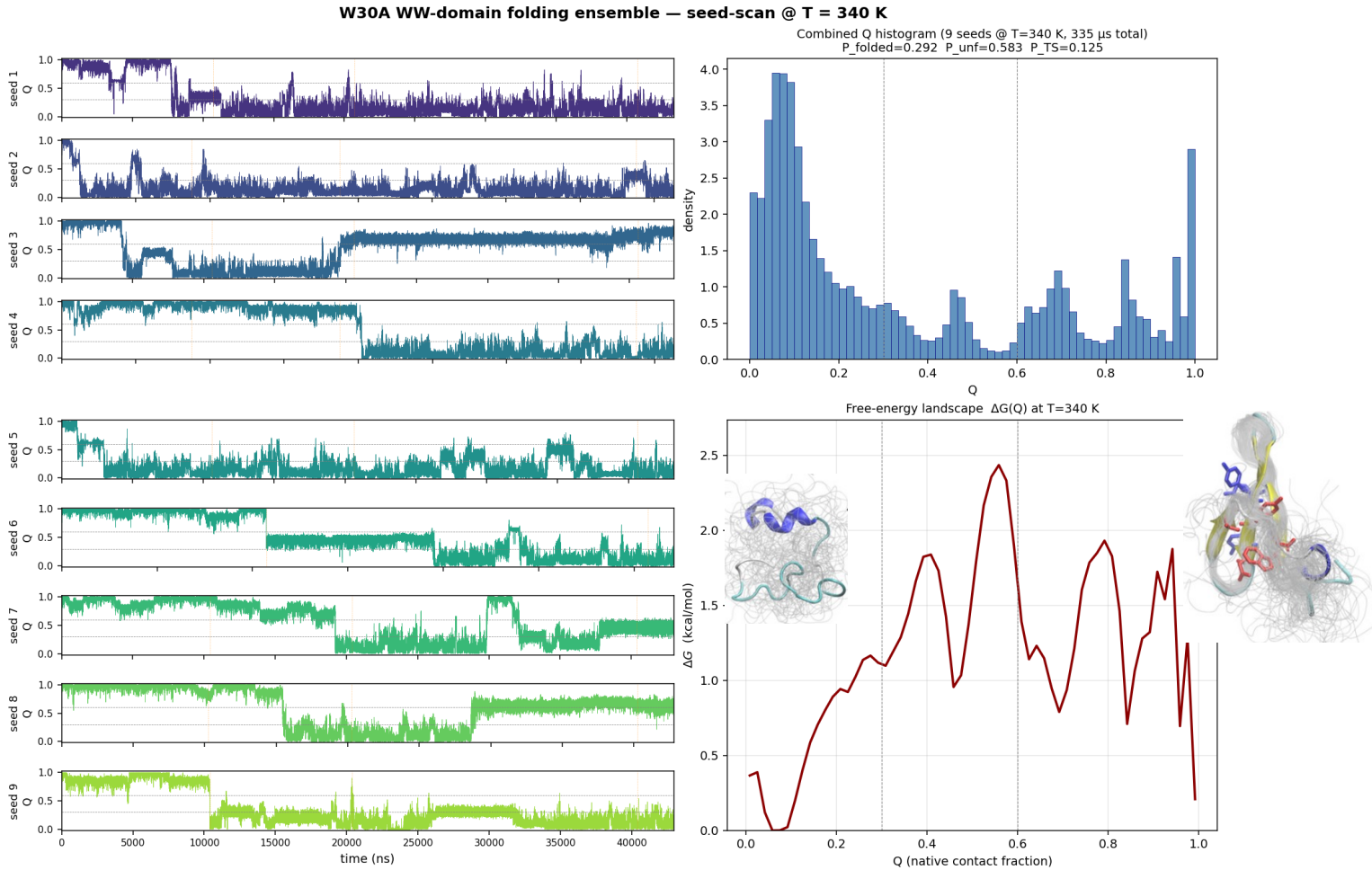


Use case 2

Finding the folding temperature of WW-domain (転移温度の探索)

Optimization

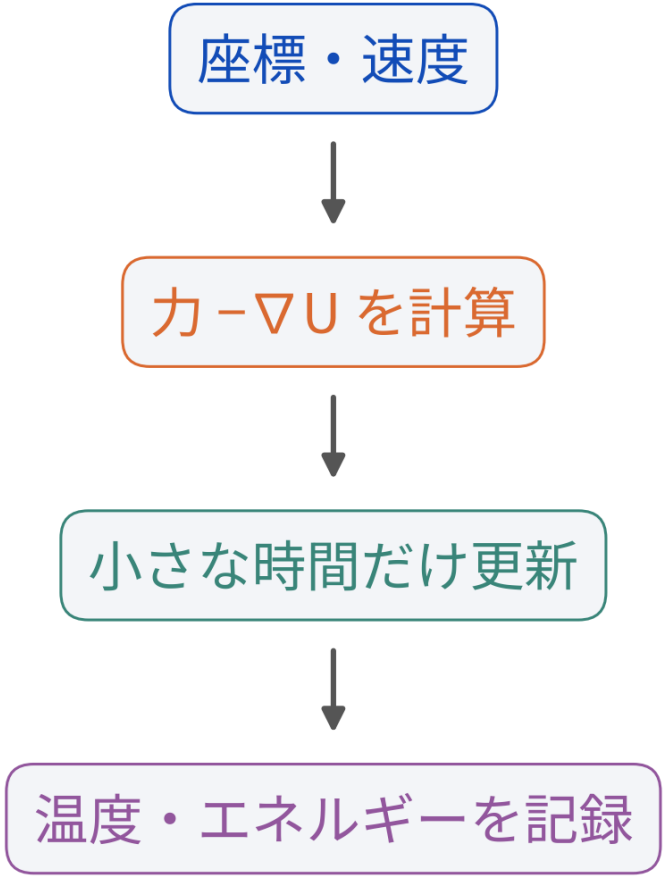
Prompt: Find the folding temperature of ww-domain by using 9 GPUs



折り畳み温度を探す利用例。温度・状態の定義、反復計算、収束の確認が必要で、一度の短い軌跡では決まらない。

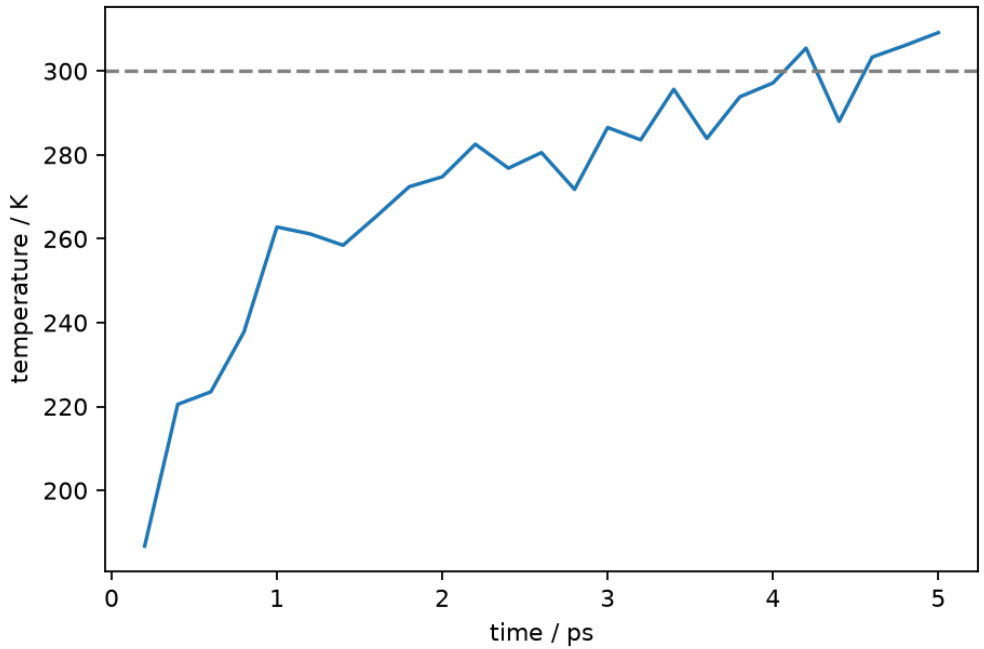
教員デモ：計算の入力から根拠までをたどる

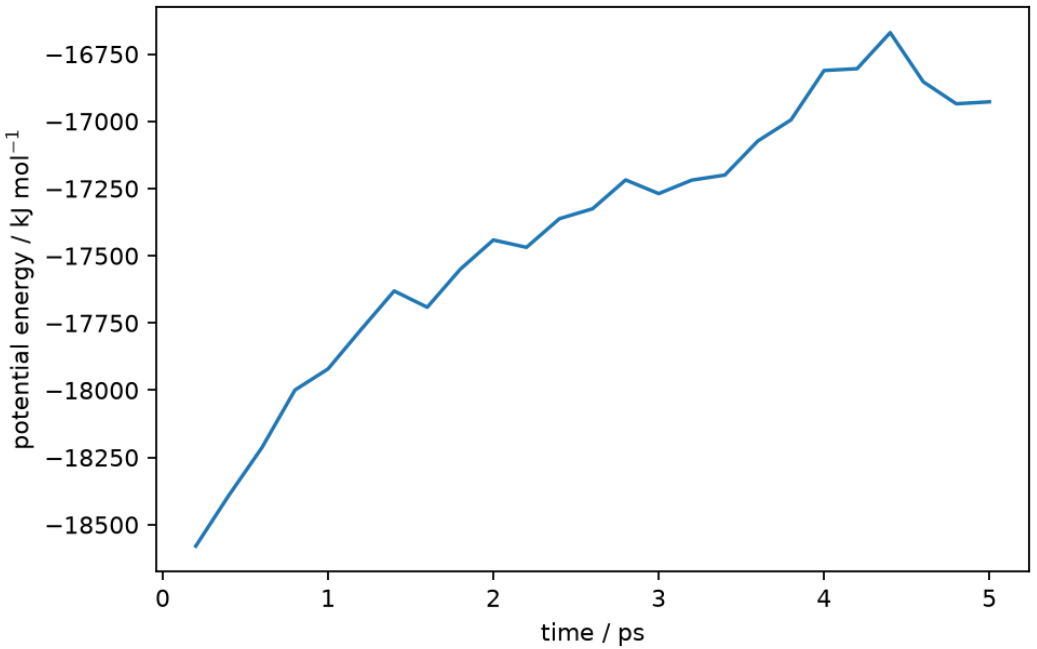
題材は第11回と同じ水中ペプチドの短い計算。
最初に条件・出力を指定した依頼文を読み、どの段階に対応するかを確認する。
次に保存された構造・温度・エネルギー・実行条件を開く。
新しい大規模計算ではなく、既存の短い実行の記録とMDClawのDAGの読み方を示す。



デモで見るもの：説明と成果物が一致するか

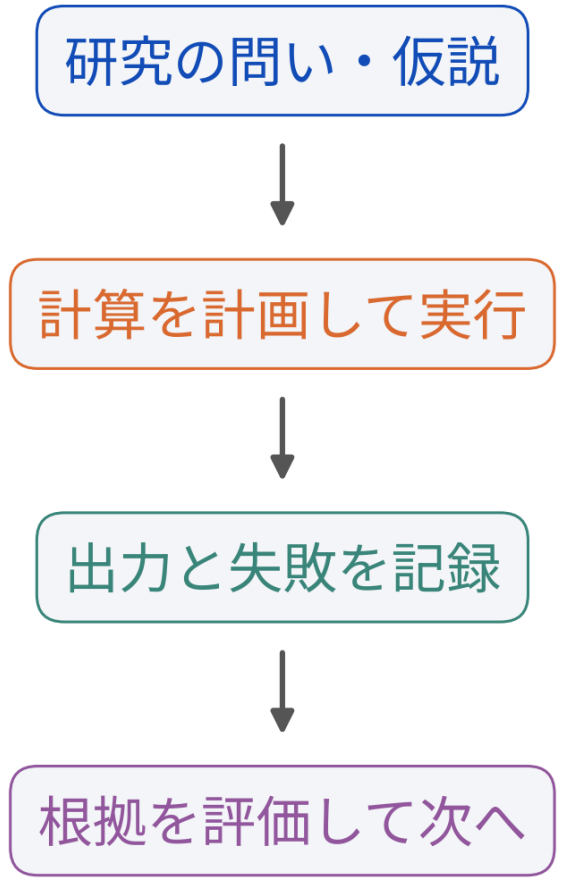
「300 Kで動かした」という説明なら、積分器の設定と温度ログを見る。
「5 ps」という説明なら、2 fs×2500ステップとの一致を見る。
「構造が安定」という説明には、計算長・反復・観測量の根拠が必要。
右は第11回Notebookを実行して得た温度・エネルギー。





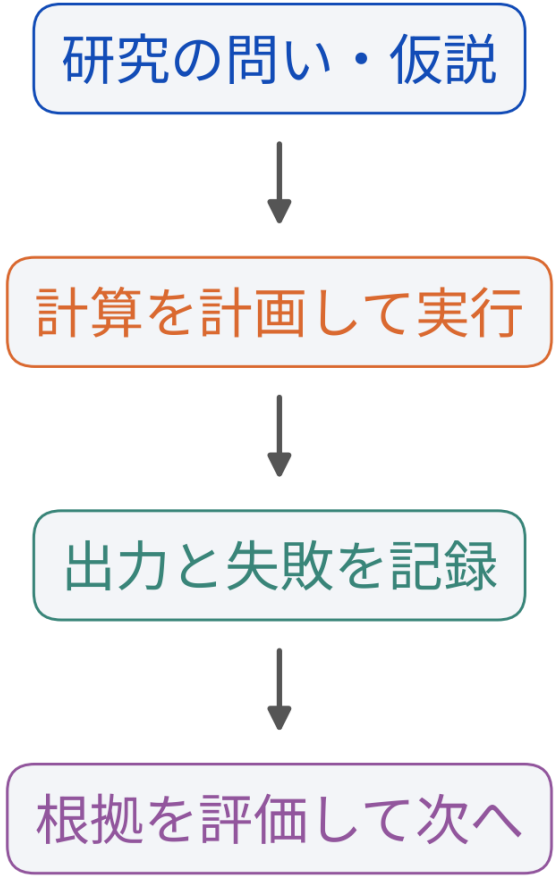
次の計算は、残っている不確実性から決める

温度は合うが構造が動かない：単に短い可能性がある。
反復ごとの割合が違う：探索や緩和が足りない可能性がある。
モデル間で結果が違う：力場や観測モデルの仮定を点検する。
追加するデータが、どの不確実性を減らすかを先に決める。



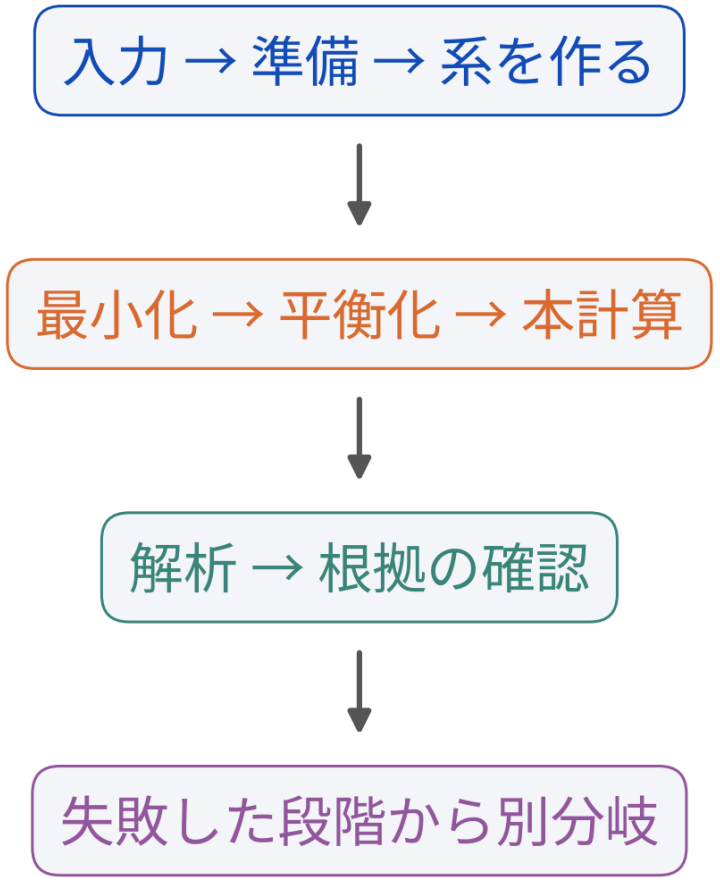
授業全体：配列・構造・分布・観測・判断を結ぶ

配列から構造を予測し、目的の構造に合う配列を設計する。
相互作用から構造の分布と動きを計算する。
実験の観測を通じてモデルを評価・補正する。
エージェントはこれらの計算と記録をつなぐ。判断の根拠は、条件のそろったデータと検証に置く。



まとめ：自動化するほど、根拠を追える記録が要る

AI Scientistは仮説や手順を広げる。計算・実験が検証の手掛かりを与える。
転移学習ではデータ数に加え、配列の多様性とラベルの意味が重要。
MDClawでは計画・操作・成果物・失敗をつなぎ、同じ条件で比較できる形に残す。
実行成功を、そのまま科学的結論へ置き換えない。



付録：導出と補足資料

ここからは本編の補足です。
導出の途中式、計算条件、研究例の詳細を示します。
本編の各手法の前提と対応させて読んでください。

MDPrepBench tasks P01-P13.

P01	**Simple monomer**	2LZM; clean one protein chain for explicit-water MD.
P02	**Chain + ligand**	1AKE; include chain A and AP5 despite chain ambiguity.
P03	**Ligand pose**	181L; preserve T4L L99A plus benzene crystal pose.
P04	**Ligand filtering**	3PWB; keep requested ligands, exclude irrelevant heterogens.
P05	**Charged cofactor**	1DAP; retain and document NDP/NADPH-like cofactors.
P06	**Supported ions**	1CLL; retain four Ca2+ ions.
P07	**Ion triage**	4RBQ; prepare RNA while preserving designated K+ ions.
P08	**Point mutation**	2LZM; branch WT to L99A without numbering drift.
P09	**Multi-mutant**	2LZM; apply L99A and M102Q together.
P10	**Disulfides**	5PTI; record BPTI disulfides and exclude deuterium.
P11	**Protonation**	2LZM; preserve explicit A:11 GLH protonation.
P12	**Deposited PTM**	5K9P; restore deposited SEP and source record.
P13	**Requested PTM**	1UBQ; convert Ser20 to SEP on request.

MDPrepBench tasks P14-P25.

P14	**Glycan**	6YA2; preserve N-linked glycans as glycans.
P15	**DNA**	5MVQ; prepare DNA without protein defaults.
P16	**RNA**	4RBQ; prepare RNA with RNA-compatible metadata.
P17	**DNA duplex**	1BNA; preserve both DNA chains and neutralization record.
P18	**Membrane**	2LOP; honor POPC:POPE:CHL1 = 2:1:1.
P19	**Candidate selection**	2K39; select specified NMR model before prep.
P20	**Terminal capping**	5AWL; honor ACE/NME caps.
P21	**PDB cleanup**	4Q5T; handle MSE, altloc, numbering, missing residues.
P22	**FF/water fidelity**	2LZM; honor ff19SB plus OPC.
P23	**Implicit solvent**	5AWL; avoid explicit water when implicit is requested.
P24	**Biological assembly**	1STP / 2MS2; generate assembly 1 and map chains.
P25	**KCl concentration**	5AWL; honor 0.30 M KCl and neutrality.

MDPrepBench: Example of a prompt (P01)

Use this prompt as the task statement. Retrieve public sources as needed, and do not read `truth/` or `scorer/` if those directories exist.

Task: Simple monomer preparation: retrieve T4 lysozyme chain A, clean it, prepare an explicit-water-compatible topology, and report that no unintended ligands were retained.

Public source anchors: PDB 2LZM.

Your submission directory must contain:

- `manifest.json`
- `metrics.json`
- `provenance.json`
- `evidence_report.json`
- `prepared_structure.pdb`

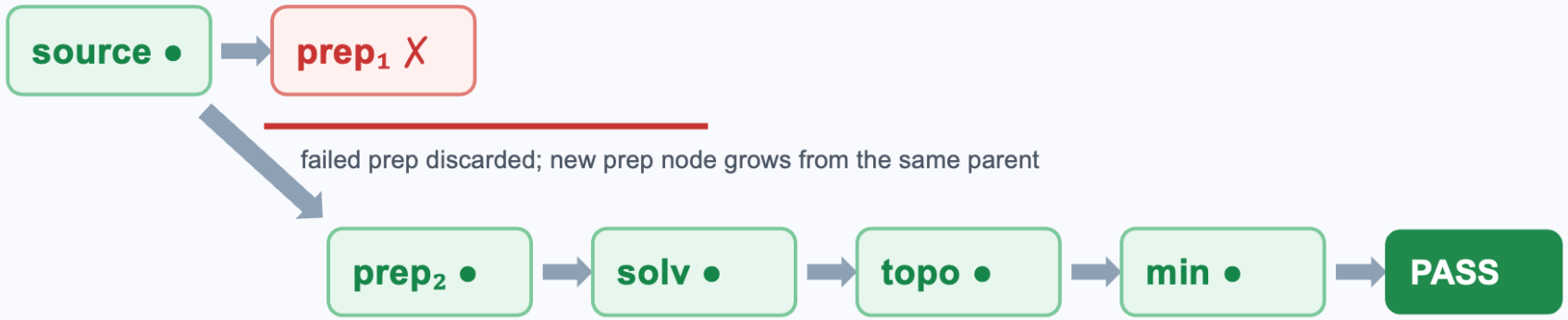
~~~~
~~~~

DAG-based workflow enables self-repair: readable examples of where the mechanism worked

Self-repair flipped FAIL → PASS

deepseek · P04 multiple ligands 3PWB

DAG state persisted after prep failed, so the run branched to $prep_2$ and continued. Without that retry, this task would have remained a fail.



Five common ways MDClaw turns a request into a recorded MD workflow.

Direct MD workflow



Create a thin study and run one jobs/main DAG.

Question-driven plan



Translate a scientific question into jobs, observables, and criteria.

End-to-end answer



Run MD, analyze trajectories, and return evidence plus uncertainty.

WT vs mutant



Branch from prep and keep variant history explicit.

Ligand / ion setup



Preserve selected components and document chemistry.

Five advanced patterns show why DAG management matters at AI-HPC scale.

PTM branch



Restore deposited PTMs or create requested modifications.

Membrane system



Embed a protein in explicit membrane and water.

Surrogate ensemble



Use BioEmu or Boltz candidates as source bundle members.

HPC continuation



Extend production segments and keep the chain checkable.

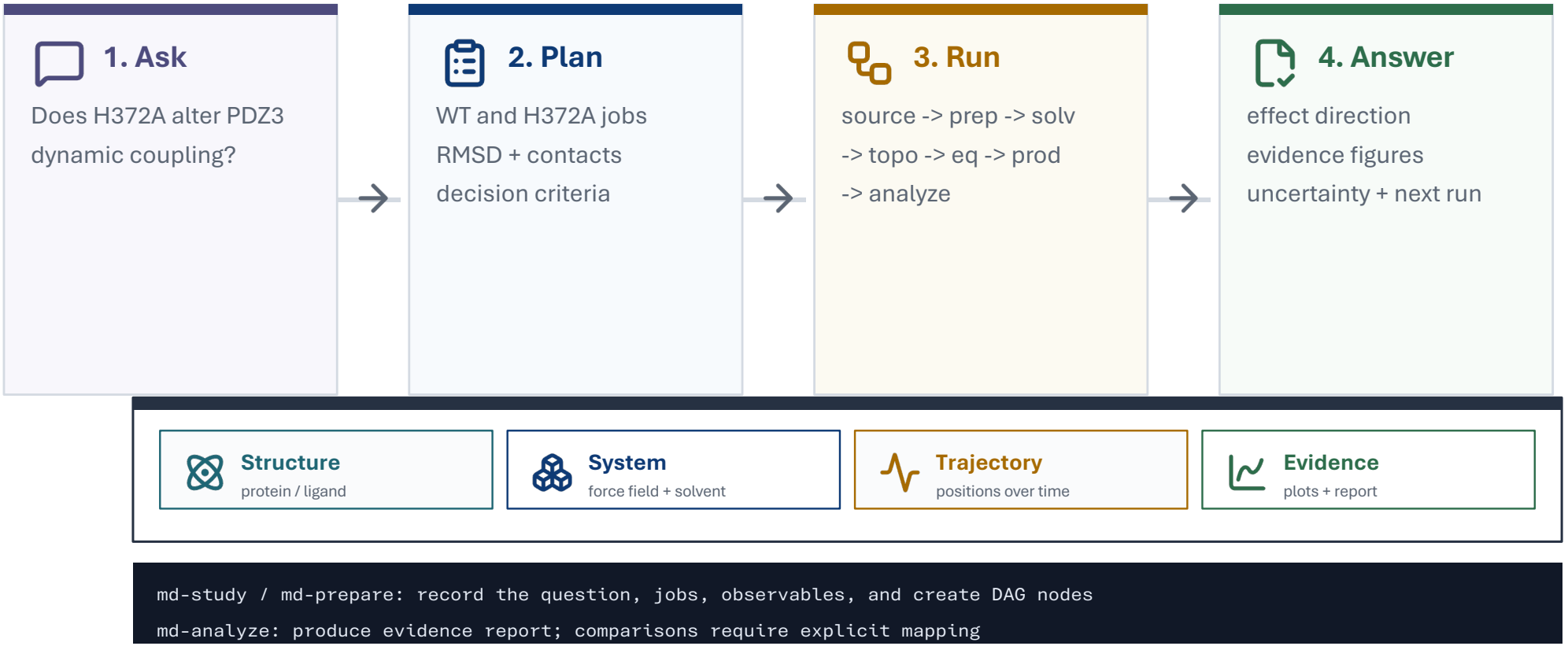
Benchmark run



Expose prompt.md, collect submissions, write score.json.

EXECUTION EXAMPLE

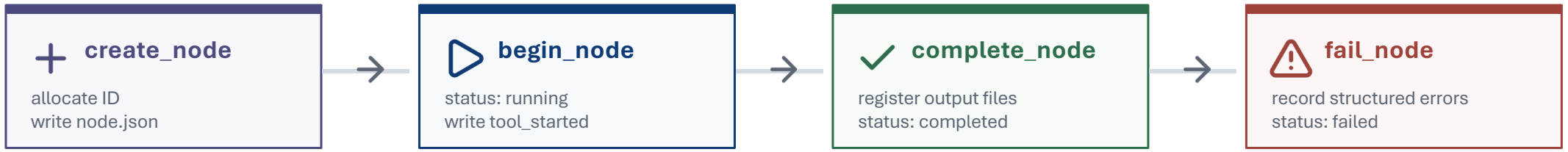
A scientific question becomes an MD plan, a DAG, and a structure-based answer.



The answer is not just prose. It is backed by output files and a recorded workflow history.

NODE LIFECYCLE

Each workflow step is created, run, completed or failed, and then kept as a record.



Sealing rule

- Completed node.json records are scientific evidence records.
- Post-completion observations go to append-only events.

Retry rule

- Same conditions can retry the same node as a new Attempt.
- Changed conditions require a new node or branch.

DATA MODEL

These names keep the scientific question separate from the run details.

Study

scientific question

MD goal

planned jobs

analysis intent

Job

one source-rooted

workflow DAG

usually jobs/main

Node

source prep solv topo

eq prod analyze

node.json + output files

Attempt

one retry of the

same node conditions

history in events

Combine MD workflows with surrogate models to answer faster.

Main idea

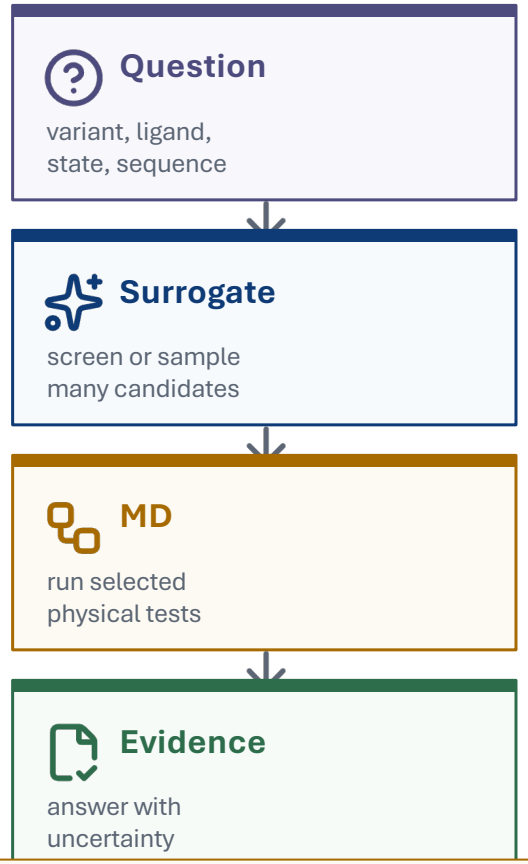
Call a surrogate model inside the MDClaw DAG before expensive MD.

Use the surrogate result to choose candidates, starting structures, or regions that deserve MD.

Why it matters

The goal is not to replace MD.

The goal is to spend MD time on the most informative simulations.



Possible outputs

- ranked structures
- candidate ensembles
- states for MD restart
- regions that need sampling
- early uncertainty estimate

Future target: use surrogate models for fast exploration, then use MD for physics-based checking of the selected cases.

FUTURE PERSPECTIVE 2

Use the DAG to search better sampling strategies and force-field choices.

Adaptive restraints with OpenMM TorchForce

Use flexible PyTorch-based coordinate restraints inside OpenMM.

Let the agentic loop test where restraints help sampling.

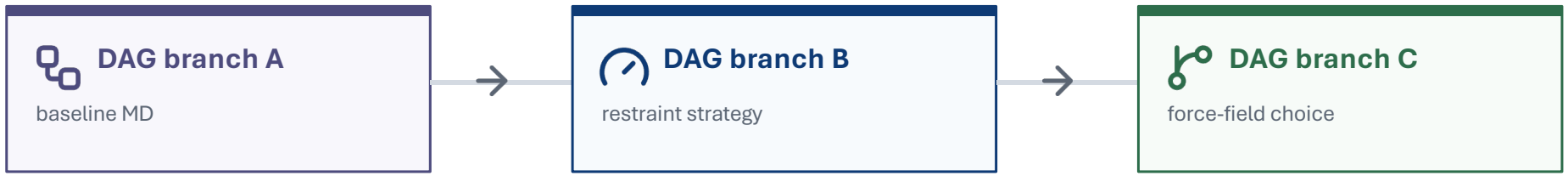
Record restraint definitions, parameters, outputs, and decisions as DAG branches.

Force-field comparison and optimization

Branch the same system across force fields, water models, ion models, or parameter sets.

Compare outputs under the same analysis plan.

Use the result to guide force-field selection or parameter optimization.



The same DAG design can support systematic sampling tests, force-field comparison, and parameter optimization.

CLIs and MCP servers are composed of open-source HPC ecosystem

Main point

The agent does not directly manage all low-level MD details. It calls MDClaw tools, and those tools call established scientific software.

Why this helps

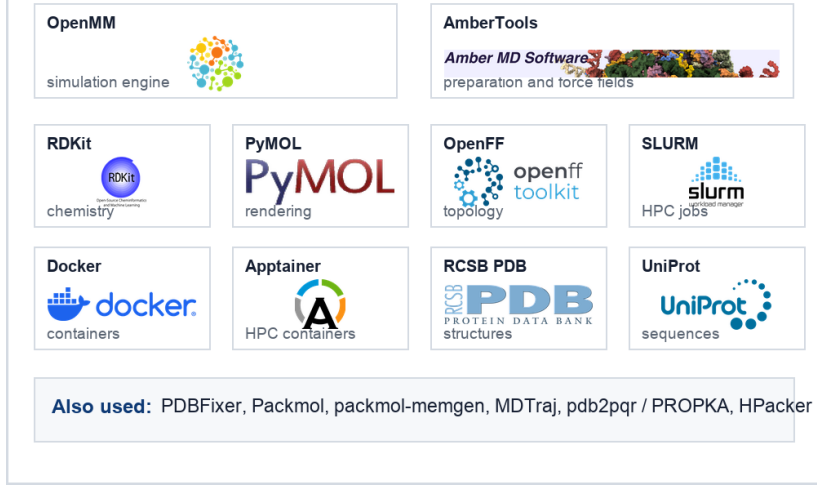
We can keep using trusted MD engines, while adding workflow records, checks, branching, and reports around them.

Scientific software stack

Input / knowledge	RCSB PDB, AlphaFold, UniProt, PubChem, PubMed
Structure preparation	PDBFixer, pdb2pqr, PROPKA, RDKit, HPacker, MODELLER / Boltz-2 / BioEmu
Solvation / membranes	Packmol, packmol-memgen, explicit water, ions, lipid composition
Force fields / topology	AmberTools, OpenFF Pablo, OpenMM XML, ff19SB, GAFF, lipid21
Simulation / analysis	OpenMM, MDTraj, PyMOL, equilibration, production, evidence reports
Execution	conda, Docker, Apptainer / Singularity, SLURM jobs and arrays

Software ecosystem

actual tools called by the workflow



Guardrails: CLI stops risky chemistry choices instead of quietly continuing.

Structured guardrail result

- field
- message
- severity
- actual / expected
- suggested_fix
- stable code

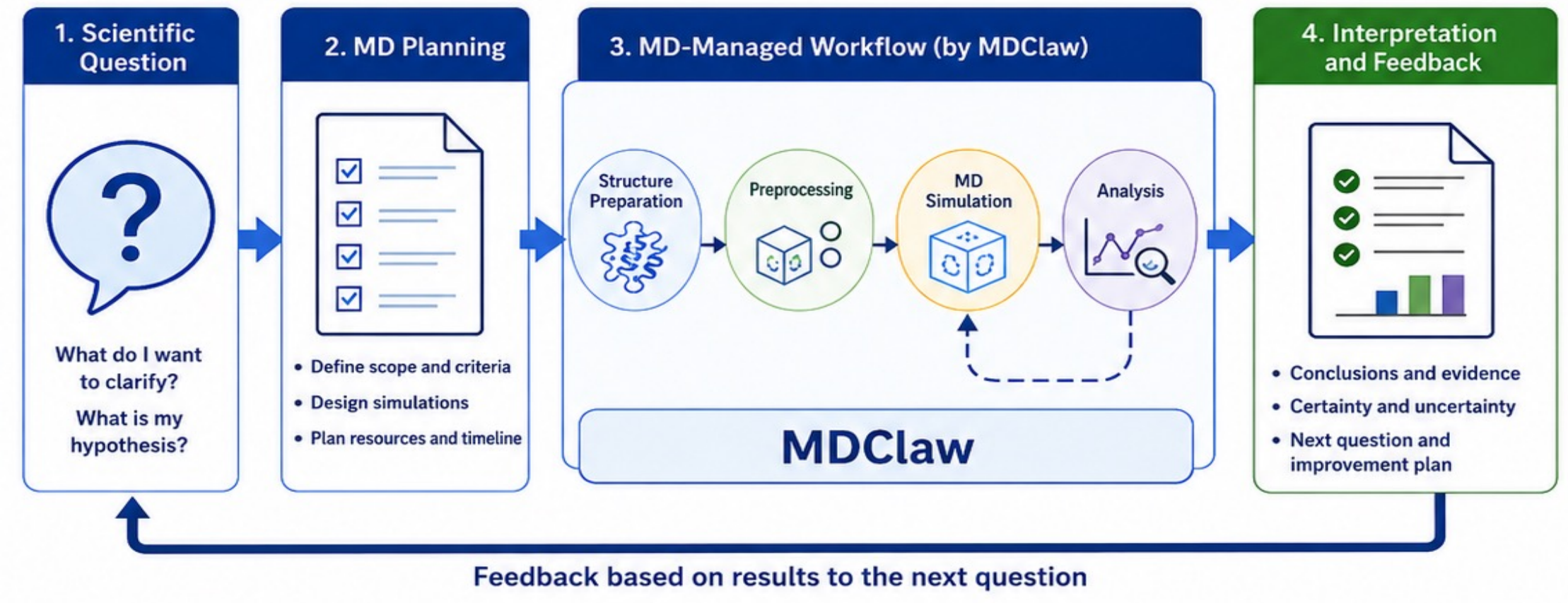
```
code: forcefield_water_blocked
actual: ff19SB + tip3p
expected: ff19SB + OPC
```

Topology contract

- prep supplies chemistry and records
- topo resolves force fields and parameters
- topo emits an OpenMM XML triple
- eq/prod/analyze consume DAG output files

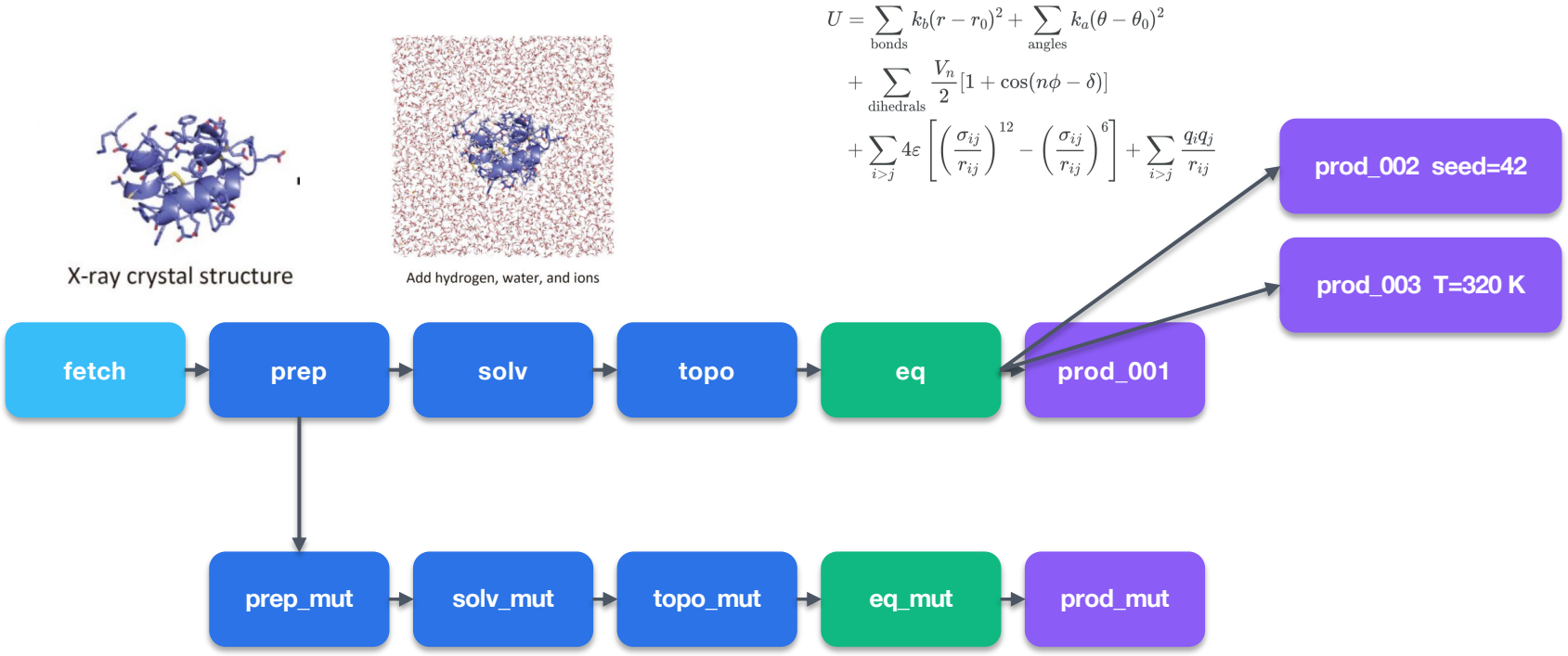
```
system.system.xml
system.topology.pdb
system.state.xml
```

Directed Acyclic Graph (DAG)-based MD workflow

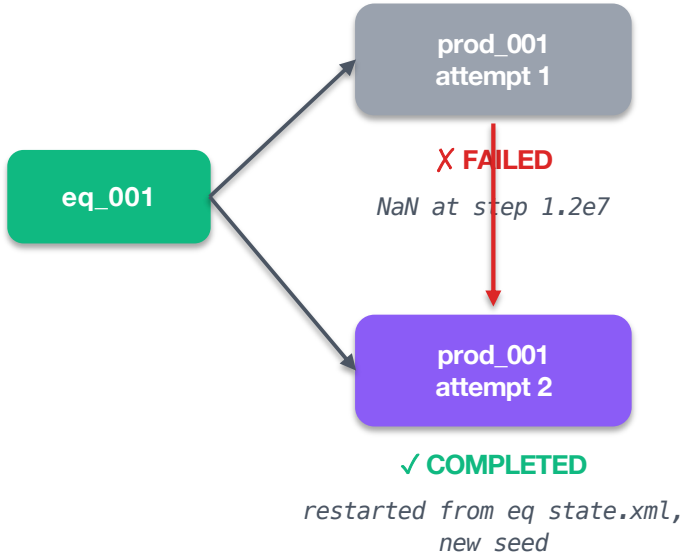


Sources: user-provided DAG.png; mdclaw/docs/developer/architecture.md; mdclaw/_node.py

DAG-based MD workflow is auditable, durable, and flexible



DAG-based MD workflow is auditable, durable, and flexible



原著・原資料と日付

AI Scientist：Robin DOI:10.1038/s41586-026-10652-y／Co-Scientist DOI:10.1038/s41586-026-10644-y／ERA DOI:10.1038/s41586-026-10658-6。
転移学習：Murakami et al. DOI:10.64898/2026.07.30.741744。図は投稿原稿Fig. 1–3。
MDClaw：github.com/matsunagalab/mdclaw。6月9日理研・6月17日蛋白質科学会の発表を再利用。
現在の役割分担は2026年10月11日のローカル実装・READMEと照合。